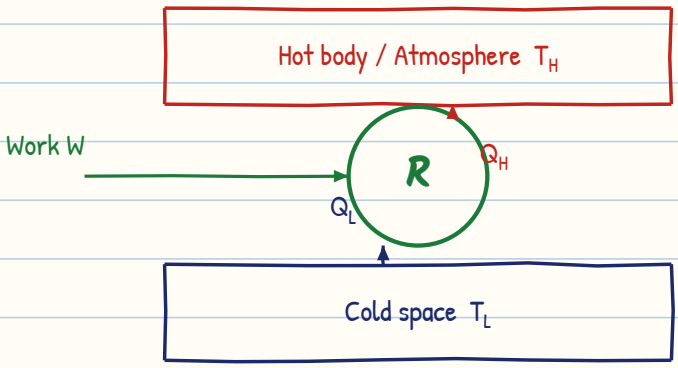
Syllabus: Necessity and applications, Carnot refrigerator, first and second law applied to refrigerating machines, unit of refrigeration, COP, EER, different refrigeration methods. Air refrigeration: Bell-Coleman cycle, ideal and actual cycles, open and dense air systems, numericals, refrigeration needs of aircraft.

I. Introduction to Refrigeration

- **Refrigeration** = the process of removing heat from a space or substance so that its temperature is lowered below that of the surroundings, and then keeping it there.
- It needs a device (refrigerating machine) which takes in **work** (or heat) and pumps heat from a low-temperature body to a high-temperature body. This is not natural: heat flows hot → cold by itself. **Second law** says we must spend work.
- **Refrigerator** → aim: extract Q_L from the cold space. **Heat pump** → aim: deliver Q_H to the warm space. Same machine, different purpose.



$Q_H = Q_L + W$

$COP = Q_L / W$

Necessity of refrigeration

- Preservation of food (milk, meat, fish, fruits, vegetables) – slows bacterial growth and enzyme action, so shelf life increases.
- Comfort for human beings: air conditioning of homes, offices, theatres, hospitals, vehicles.
- Industrial process control: machine tools, precision measuring rooms, textile (humidity control), printing, photography film.
- Storage of medicines, vaccines, blood banks (cold chain).

Applications

- **Domestic:** refrigerator, deep freezer, water cooler, room AC.

- **Commercial:** cold storage, ice plant, hotels, supermarkets, display cabinets.
- **Industrial:** chemical industry (separation of gases, liquefaction of O_2, N_2, LPG), petroleum dewaxing, steel (cooling of blast furnace air), dairy, brewery.
- **Transport:** refrigerated trucks, rail vans, ships, aircraft cabin cooling.
- **Others:** cooling concrete of dams, mining (cooling mines), ice skating rinks, cryogenics, space research, superconductors.

2. First and Second Law applied to Refrigerating Machines

First law (energy conservation)

For a cyclic device, net heat = net work. Taking the refrigerator as the system (heat out is +ve):

$$Q_H = Q_L + W$$

where Q_L = heat absorbed from cold space, Q_H = heat rejected to hot body, W = work input.

Second law

- **Clausius statement:** It is impossible for a self-acting machine, unaided by external agency, to move heat from a body at lower temperature to one at higher temperature.
- **Kelvin-Planck statement:** It is impossible to construct a device operating in a cycle that produces net work while exchanging heat with a single reservoir.
- Meaning for refrigerator: W cannot be zero, hence COP is always finite. A refrigerator with $W = 0$ (a "perpetual motion machine of second kind") is impossible.
- The two statements are equivalent: violation of one means violation of the other.

Coefficient of Performance (COP)

$$COP_R = \text{Refrigerating effect} / \text{Work input} = Q_L / W = Q_L / (Q_H - Q_L)$$

$$COP_{HP} = Q_H / W = Q_H / (Q_H - Q_L) \; ; \; COP_{HP} = COP_R + 1$$

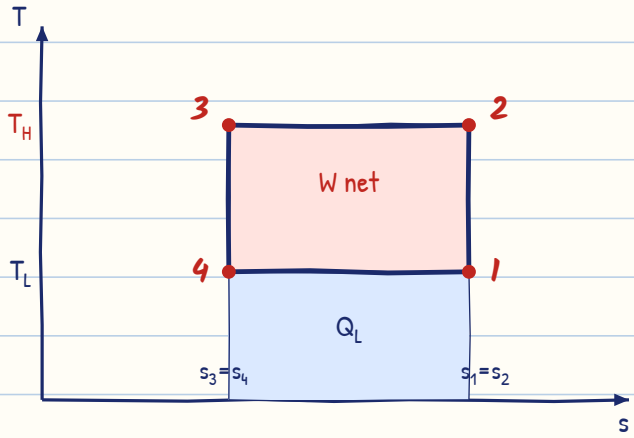
Note: COP of a refrigerator can be greater than 1 (it is not an efficiency, since it is output/input of different forms of energy). COP of heat pump is always more than 1.

Heat pump (reversed cycle) – quick idea

- The same cycle used to heat a space: Q_H is the useful effect. $COP_{HP} = Q_H/W = COP_R + 1 > 1$.
- Example: a refrigerator of COP 3 used as heat pump gives COP 4, so 1 kW of electricity delivers 4 kW of heat – much better than a 1 kW heater.
- Carnot heat pump between 5 °C and 35 °C: COP = 308/30 = **10.27**.

3. Carnot Refrigerator (Reversed Carnot Cycle)

The Carnot cycle is the most efficient reversible cycle between two temperatures T_H and T_L. Reversing the Carnot engine gives the Carnot refrigerator.



1-2 Isentropic compression ($T_L \rightarrow T_H$)

2-3 Isothermal heat rejection at T_H (Q_H)

3-4 Isentropic expansion ($T_H \rightarrow T_L$)

4-1 Isothermal heat absorption at T_L (Q_L)

Area 4-1-s-s = Q_L, Area 1-2-3-4 = W

- Heat absorbed (4-1): $Q_L = T_L (s_1 - s_4)$
- Heat rejected (2-3): $Q_H = T_H (s_2 - s_3)$ with $s_1=s_2$, $s_3=s_4$
- Net work: $W = Q_H - Q_L = (T_H - T_L)(s_1 - s_4)$

$$COP_{Carnot} = T_L / (T_H - T_L) \; ; \; COP_{HP,Carnot} = T_H / (T_H - T_L)$$

Observations (important for exams)

- COP depends **only on temperatures** of the reservoirs, not on the working fluid.
- COP rises when T_L is raised or T_H is lowered (i.e. smaller temperature lift).
- No actual refrigerator can beat Carnot COP between the same temperatures. It is the **upper limit** and the yard-stick for comparison. **Relative COP = Actual COP / Carnot COP**

Why Carnot cycle is not used practically

- Isothermal heat transfer needs infinitely slow processes; compression of a wet vapour (state 1) is difficult – liquid droplets damage the compressor (wet compression).

- Isentropic expansion 3-4 (a work-recovering expander) of liquid-vapour mixture gives little work – not worth the cost; replaced by a throttle valve in VCR.
- For gases (air) isothermal processes are impossible in a machine → Bell-Coleman cycle.

4. Unit of Refrigeration

- The capacity of a refrigerating machine is measured in **Tonne of Refrigeration (TR)**.
- 1 TR = the heat to be removed from 1 tonne (2000 lb, short ton) of water at 0 °C to form ice at 0 °C in 24 hours.

$$1\ TR = 2000 \times 144\ BTU / 24\ h = 12\ 000\ BTU/h = 200\ BTU/min$$

$$1\ TR = 211\ kJ/min = 3.5167\ kW \approx 3.5\ kW \approx 3024\ kcal/h$$

- Latent heat of ice = 335 kJ/kg (144 BTU/lb). 1 BTU = 1.055 kJ.
- Other units: 1 kW = 3412 BTU/h = 860 kcal/h. Domestic fridge ≈ 0.1 TR; window AC 1-2 TR.

EER and other performance indices

- **EER** (Energy Efficiency Ratio) = cooling capacity in BTU/h divided by electrical power input in Watts.

$$EER = Q_L(BTU/h) / P(W)\ ;\ EER = 3.412 \times COP$$

- **SEER** = seasonal EER, the cooling delivered over a whole season / energy consumed (more realistic). **ISEER** in India.
- Another measure: kW/TR = 3.5167 / COP (lower is better). For chillers a good value is 0.6 kW/TR.

Index	Unit	Typical value
COP	dimensionless	2.5 - 5 (air-conditioner)
EER	BTU/h per W	9 - 12
kW/TR	kW per TR	0.6 - 1.2
SEER	BTU/h per W	13 - 22

5. Different Methods of Refrigeration

Refrigeration methods are classified into **non-cyclic** (one-time cooling) and **cyclic** (continuous).

Non-cyclic methods

- **Ice refrigeration:** ice melts at 0 °C absorbing 335 kJ/kg; used in fish and vegetable transport. Cheap but temperature limited to ≈ 0 °C.
- **Ice + salt (freezing mixtures):** gives temperature below 0 °C (NaCl: −21 °C, $CaCl_2$: −50 °C).
- **Dry ice (solid CO_2):** sublimes at −78.5 °C at 1 atm, latent heat 573 kJ/kg; used for ice cream and transport of frozen goods.
- **Evaporative cooling:** water evaporates and absorbs latent heat from air (desert cooler); works only in dry climate.

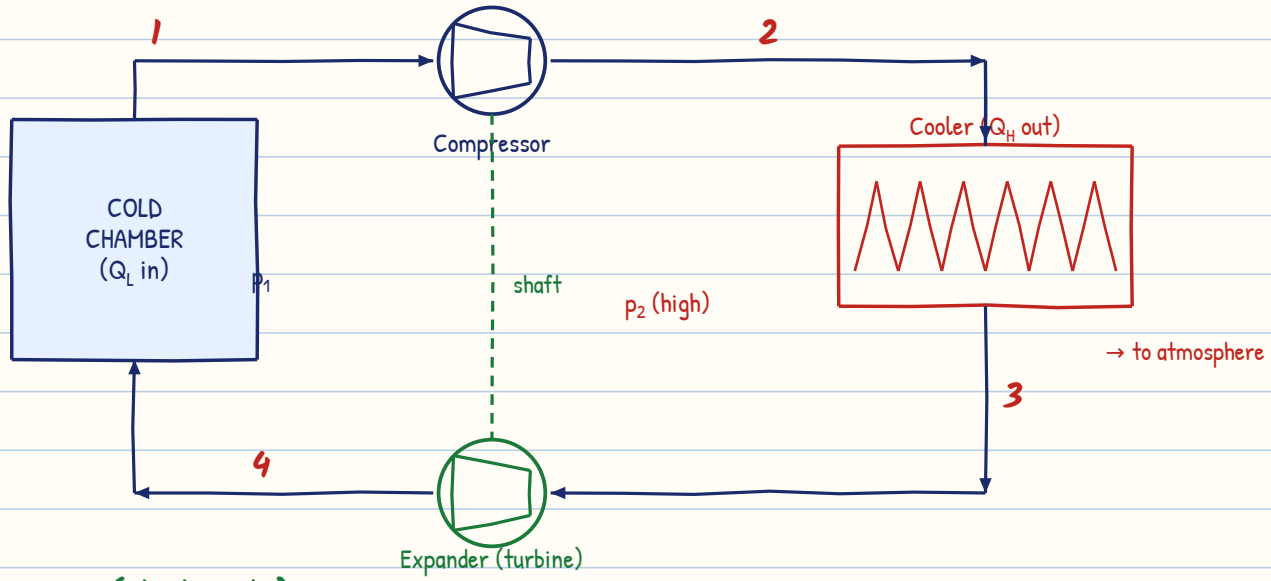
Cyclic methods

Method	Working fluid	Energy input
Air refrigeration (Bell-Coleman)	Air	Mechanical work
Vapour compression (VCR)	Freon / NH_3	Mechanical work
Vapour absorption (VAR)	NH_3-water, LiBr-water	Heat
Steam jet	Water vapour	High pressure steam
Thermo-electric (Peltier)	Electrons in junction	DC electricity
Vortex tube (Hilsch)	Compressed air	Pressure energy
Magnetic (adiabatic demag.)	Paramagnetic salt	Magnetic field

- Cryogenic (gas liquefaction – Linde, Claude) methods reach below −150 °C.

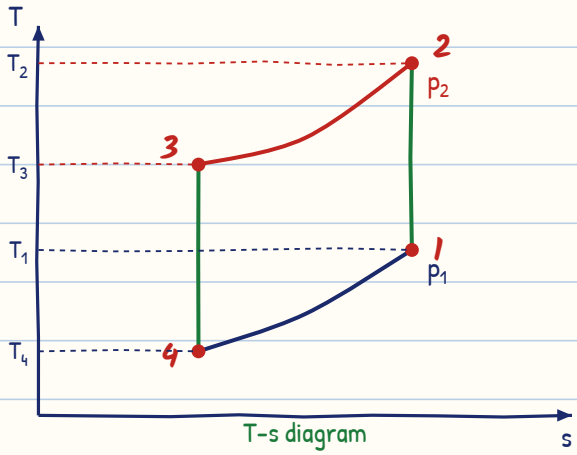
6. Air Refrigeration – Bell-Coleman (Reversed Joule / Brayton) Cycle

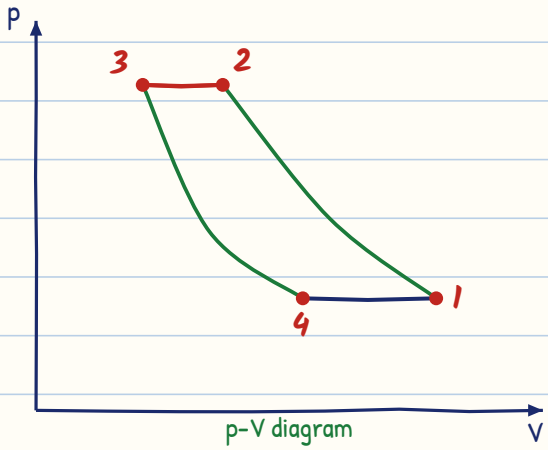
- Working fluid is **air** (a gas, no phase change). Also called the **reversed Brayton** or **Joule cycle**. Used in aircraft cooling and gas liquefaction.
- Heat rejection and absorption happen at **constant pressure**, not constant temperature (a gas cannot absorb heat isothermally in a heat exchanger).



Processes (ideal cycle)

- **1-2** Isentropic compression of air from cold chamber (p_1, T_1) to (p_2, T_2) – work input.
- **2-3** Constant pressure cooling in cooler – heat Q_H rejected to the atmosphere, $T_3 > T_0$.
- **3-4** Isentropic expansion in expander (turbine) to p_1; temperature falls to T_4 T_1; work is recovered.
- **4-1** Constant pressure heating in cold chamber – heat Q_L absorbed from the refrigerated space.





Cycle analysis

Take unit mass of air, c_p = 1.005 kJ/kg K, γ = 1.4.

$$Q_L = c_p(T_1 - T_4) \quad Q_H = c_p(T_2 - T_3)$$

$$W_{net} = W_c - W_e = c_p[(T_2 - T_1) - (T_3 - T_4)]$$

$$COP = (T_1 - T_4) / [(T_2 - T_1) - (T_3 - T_4)]$$

For isentropic processes 1-2 and 3-4: $T_2/T_1 = T_3/T_4 = (p_2/p_1)^{(\gamma-1)/\gamma} = r_p^{(\gamma-1)/\gamma}$

$$COP = 1 / [\, r_p^{(\gamma-1)/\gamma} - 1 \,] = T_4 / (T_3 - T_4) = T_1 / (T_2 - T_1)$$

Remarks

- COP **decreases** with increase of pressure ratio r_p. (but a larger r_p gives lower T_4, more cooling per kg).
- COP of Bell-Coleman < Carnot COP between T_1 and T_3 (because the cycle works over a wider temperature range T_4 to T_2).
- Refrigerating effect is small (cp of air low) → large mass flow, bulky machine. Hence low COP, used only where air is needed (aircraft).

Merits and demerits of air refrigeration

- **Merits:** air is cheap, non-toxic, non-flammable, no leakage problem; light and simple for aircraft; can give very low temperatures (−60 °C or less) by increasing r_p.
- **Demerits:** low COP (≈ 0.5 - 2); large mass flow, bulky equipment; moisture in air freezes and chokes the expander; higher running cost; power needed is high.

Item	Air cycle	Vapour compression
Working fluid	Air (gas)	Refrigerant (changes phase)
Heat transfer	Constant pressure, sensible	Constant T & p, latent
COP	Low (≈ 1)	High (3 - 5)
Equipment size	Large	Compact
Main use	Aircraft, liquefaction	Homes, industry, cold storage

Actual Bell-Coleman cycle

- Real compression is not isentropic: state 2 → **2'** (more work). Real expansion gives 4 → **4'** (higher T, less cooling).
- Pressure drops in coolers, pipes and cold chamber; heat leakage.

$$\eta_c = (T_2 - T_1) / (T_2' - T_1) \quad \eta_e = (T_3 - T_4') / (T_3 - T_4)$$

$$COP_{actual} = (T_1 - T_4') / [(T_2' - T_1) - (T_3 - T_4')]$$

Regenerative air cooling (gas refrigeration with regenerator)

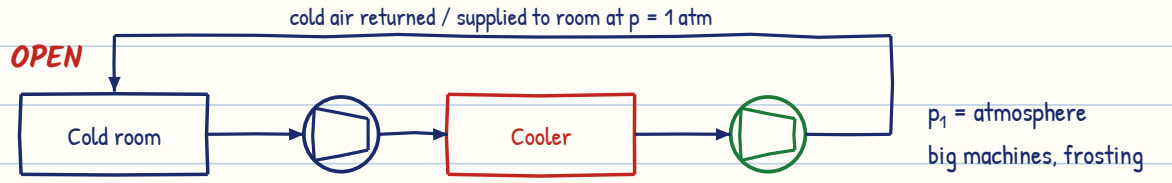
- Warm air leaving the cooler (state 3) is further cooled in a **regenerator** by the cold air coming from the cold chamber, before entering the expander. The expander inlet becomes $T_3' < T_3$, so lower exit temperature T_4 is reached with the **same pressure ratio**.
- Cold air leaving the cold chamber is pre-heated in the regenerator before compression: compressor work rises a little, but the refrigeration effect rises much more.
- Used in air liquefaction (Linde-Hampson, Claude) and in supersonic aircraft. Energy balance of regenerator: $T_3 - T_3' = T_1' - T_1$.

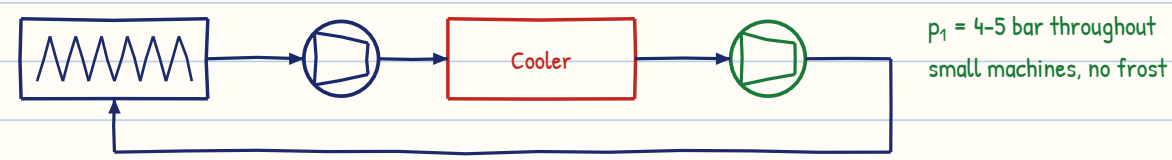
$$Q_L = c_p(T_1 - T_4) \;,\; W = c_p[(T_2 - T_1') - (T_3' - T_4)]$$

Points to remember while solving air-cycle numericals

- Always convert temperatures to Kelvin and pressures to the same unit (bar or kPa).
- Find T_2 and T_4 using the isentropic relation first, then correct with η_c and η_e.
- Refrigerating effect = $c_p(T_1 - T_4)$ per kg; mass flow = capacity (kW)/effect.
- Power = $m \times W_{net}$; COP = capacity / power. Compare with Carnot COP to show relative COP.

Open and Dense air systems





Open air system	Dense (closed) air system
Air from cold chamber is compressed, cooled, expanded and then **discharged to the cold chamber** (or fresh air from atmosphere is cooled and supplied)	Air circulates in a **closed loop** at a pressure above atmospheric throughout; the cold chamber is cooled through a heat exchanger
Lower pressure at 1 = atmospheric, so large volume of air → big machines	Higher density (hence "dense") → smaller compressor & expander for the same capacity
Moisture in air freezes in the expander → choking	Air can be dried; no frosting problem
Low COP	Higher COP due to lower pressure ratio at same pressure level

Note: Dense air: pressure ratio is the same but the low-pressure side is kept at, say, 4-5 bar instead of 1 bar, so the pressure difference (and friction/leakage) are lower and the **size reduces**.

7. Numerical Problems

Problem 1 (ideal cycle)

A Bell-Coleman refrigerator takes air at 1 bar and 270 K (cold chamber), compresses it to 5 bar; after cooling the air leaves the cooler at 300 K. Find: (i) T_2, T_4 (ii) COP (iii) mass flow and power for 10 TR. ($\gamma = 1.4$, $c_p = 1.005$) **Solution:**

$$T_2 = 270 \times 5^{0.2857} = 427.6 \text{ K } T_4 = 300 / 5^{0.2857} = 189.4 \text{ K}$$

$$Q_L = 1.005\ (270 - 189.4) = 81.0 \text{ kJ/kg } W = 1.005\ [(427.6-270) - (300-189.4)] = 47.3 \text{ kJ/kg}$$

$$COP = 81.0 / 47.3 = 1.71$$

Capacity = 10 × 3.5167 = 35.17 kW → mass flow m = 35.17/81.0 = **0.434 kg/s** (26 kg/min); Power = m W = 0.434 × 47.3 = **20.5 kW**

Note: Carnot COP between 270 K and 300 K = 270/30 = 9, much higher than 1.71.

Problem 2 (Carnot)

A Carnot refrigerator works between −10 °C and 30 °C. Find COP and power needed for 10 TR.

$$COP = 263 / (303 - 263) = 6.575 \; P = 35.167 / 6.575 = \mathbf{5.35\ kW}$$

Problem 3 (actual cycle)

Air at 263 K, 1 bar is compressed to 4 bar; cooled to 298 K, expanded back to 1 bar. η_c = 85 %, η_e = 90 %. Find COP.

$$T_2 = 263 \times 4^{0.2857} = 390.8\ K \; ; \; T_4 = 298/4^{0.2857} = 200.5\ K$$

$$T_2' = 263 + (390.8-263)/0.85 = 413.4\ K \; ; \; T_4' = 298 - 0.9(298 - 200.5) = 210.3\ K$$

$$Q_L = 1.005\ (263 - 210.3) = 53.0\ kJ/kg \; ; \; W = 1.005\ [(413.4-263) - (298-210.3)] = 63.0\ kJ/kg$$

$$COP_{actual} = 53.0 / 63.0 = \mathbf{0.84}\ (ideal = 1/(4^{0.2857}-1) = 1.97)$$

Problem 4 (unit conversion)

A cold store needs 100 kW cooling. Express in TR: 100 / 3.5167 = **28.4 TR**. If EER is 10 → electrical power = (100×3412/1000 BTU/h = 341 200 BTU/h)/10 = 34.1 kW.

Problem 5 (aircraft simple system)

An aircraft flies at 250 m/s where ambient air is 0.7 bar, 250 K. Ram air is brought to rest isentropically. Bleed air (ram state) is compressed to 4.5 bar (η_c = 80 %), cooled to the ram temperature + 20 K, expanded in a turbine to cabin pressure 1.013 bar (η_t = 85 %). Cabin is held at 298 K and load is 20 kW. Find air mass flow and COP. **Solution:**

$$T_0 = 250 + 250^2/(2\times1005) = 281.1\ K \; ; \; p_0 = 0.7\ (281.1/250)^{3.5} = 1.055\ bar$$

$$T_2 = 281.1\ (4.5/1.055)^{0.2857} = 425.4\ K \; ; \; T_2' = 281.1 + (425.4-281.1)/0.8 = 461.5\ K$$

$$T_3 = 281.1 + 20 = 301.1\ K \; ; \; T_4 = 301.1\ (1.013/4.5)^{0.2857} = 196.6\ K \; ; \; T_4' = 301.1 - 0.85\ (301.1-196.6) = 212.3\ K$$

$$m = 20 / [1.005 (298 - 212.3)] = \mathbf{0.232\ kg/s}$$

$$W_c = m\, c_p(T_2' - T_0) = 42.1\ kW\ ;\ W_t = m\, c_p(T_3 - T_4') = 20.7\ kW\ ;\ W_{net} = 21.4\ kW$$

$$COP = 20 / 21.4 = \mathbf{0.935}$$

Problem 6 (effect of pressure ratio)

For $T_1 = 270$ K, $T_3 = 300$ K, find COP for $r_p = 3$ and 5.

$$r_p = 3:\ COP = 1/(3^{0.2857} - 1) = \mathbf{2.71}\ (T_4 = 219.2\ K)$$

$$r_p = 5:\ COP = 1/(5^{0.2857} - 1) = \mathbf{1.71}\ (T_4 = 189.4\ K)$$

Note: Higher pressure ratio → colder air but lower COP. Designer must compromise.

Problem 7 (ice making)

How much power is needed to make 5 tonnes (5000 kg) of ice per day from water at 25 °C to ice at −5 °C, if the actual COP is 3.5? (c_{pw}=4.187, L=335, c_{pi}=2.1 kJ/kg K)

Heat removed per kg = 4.187×25 + 335 + 2.1×5 = 104.7 + 335 + 10.5 = 450.2 kJ/kg

$$Q_L = 5000 \times 450.2 / (24 \times 3600) = 26.05\ kW = \mathbf{7.41\ TR}$$

Power = 26.05 / 3.5 = **7.44 kW**

Problem 8 (EER)

A 1.5 TR air conditioner draws 1.6 kW. Find COP and EER.

$$Q_L = 1.5 \times 3.5167 = 5.275\ kW\ ;\ COP = 5.275/1.6 = \mathbf{3.30}\ ;\ EER = 3.412 \times 3.30 = \mathbf{11.25}$$

8. Refrigeration Needs of Aircraft

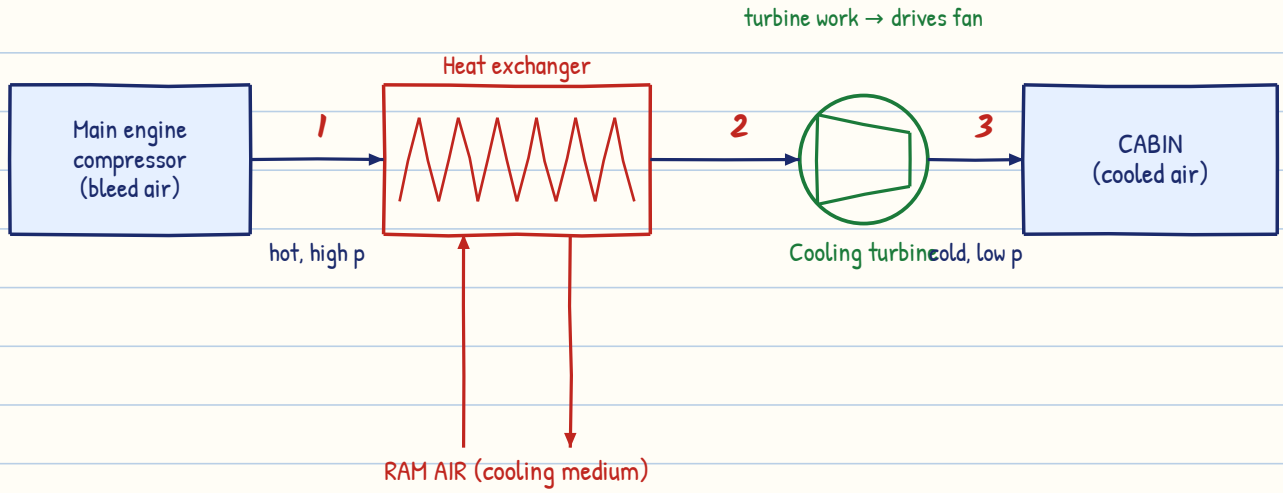
Why air conditioning of aircraft is needed

- **Solar radiation** through windows and skin.

- **Skin friction (kinetic heating)** – at high speed the air stagnates on the skin and heats it. Stagnation temperature rise: $\Delta T = V^2/(2c_p) = T_a(\gamma-1)M^2/2$. At Mach 2 it is more than 100 K.
- **Metabolic heat** of passengers and crew (~100 W each), engine heat, hot electronic/avionic equipment and lighting.
- At altitude, ambient pressure and temperature are low (−50 °C at 10 km) so cabin must be **pressurised** and kept warm in cruise but cooled on ground and at high Mach.

Why air cycle is used in aircraft

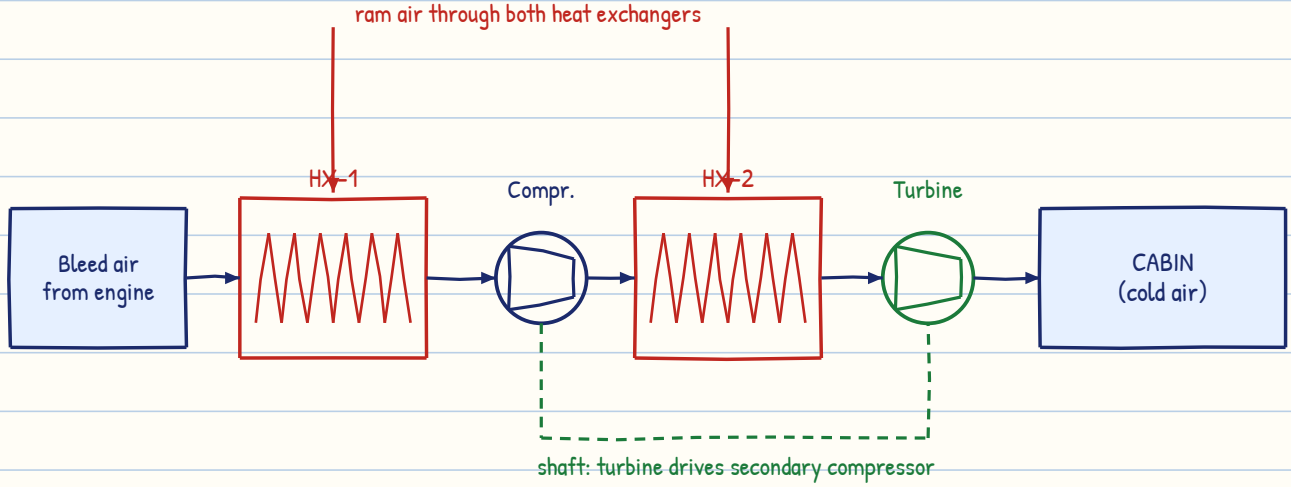
- Air is **freely available** (bleed air from engine compressor / ram air) – no refrigerant storage or leakage hazard.
- **Light weight and compact** system; small cooling turbine; high-pressure air available for free.
- Safe (non-toxic, non-flammable); simple maintenance; works at high altitude where vapour systems fail.
- Low COP does not matter, as the energy is a small part of engine power.



Types of aircraft air-cooling systems

- **Simple system:** bleed air → heat exchanger (ram air cooled) → cooling turbine → cabin (shown above).
- **Boot-strap system:** an additional compressor driven by the turbine raises pressure before the second heat exchanger → lower turbine exit temperature, used for high speed flight.

- **Regenerative system:** a regenerator cools the air before the turbine using the cold turbine exit air; for supersonic aircraft.
- **Reduced ambient system:** cooling turbine outlet is at ambient pressure reduced to altitude pressure; best for very high Mach numbers.



Ram effect

$$T_0 = T_a + V^2/(2\,c_p) = T_a[1 + (\gamma-1)M^2/2]$$

Example: $M = 0.8$, $T_a = 250$ K $\rightarrow T_0 = 250\,(1 + 0.2\times0.64) = 282$ K (rise of 32 K). Ram efficiency $\eta_r = (p_0 - p_a)/(p_0' - p_a)$ accounts for losses in the intake.

9. Quick Revision and Important Questions

- $COP_{Carnot} = T_L/(T_H - T_L)$; 1 TR = 3.5167 kW = 211 kJ/min; EER = 3.412 COP.
- Bell-Coleman COP = $1 / (r_p^{(\gamma-1)/\gamma} - 1)$; actual cycle includes η_c, η_e.
- Heat pump COP = refrigerator COP + 1.

One-page summary (mind map)

Topic	Key formula / idea
Reversed Carnot	$COP = T_L/(T_H-T_L)$; upper limit
TR	1 TR = 3.5167 kW = 211 kJ/min = 12000 BTU/h
Heat pump	$COP_{HP} = COP_R + 1$
Bell-Coleman	$COP = 1/(r_p^{0.2857} - 1)$; 4 processes: 2 isentropic + 2 constant p
Actual cycle	$\eta_c = (T_2-T_1)/(T_2'-T_1)$; $\eta_e = (T_3-T_4')/(T_3-T_4)$
Dense air	Closed loop, p_1 > atm, smaller size
Aircraft	Ram effect, bleed air, simple / boot-strap / regenerative / reduced ambient

Short answers (2 marks)

- **Define 1 TR.** Heat removal rate to freeze 1 tonne of water at 0 °C into ice at 0 °C in 24 h = 3.5167 kW = 211 kJ/min.
- **Why is COP of a refrigerator > 1?** Because it moves heat Q_L using little work W; the heat is not converted from work.
- **What is relative COP?** Ratio of actual COP to theoretical (Carnot) COP; always less than 1.
- **What is dense air system?** Closed air cycle with pressure above atmosphere on low side – smaller size, no frosting.
- **Why does aircraft use air cycle?** Light weight, air freely available, safe, high pressure bleed air is on board.
- **Ideal vs actual Bell-Coleman?** Actual has irreversible compression/expansion (η_c, η_e), pressure drops and heat leak; COP lower.
- **Define EER.** BTU/h of cooling per watt of electric input; EER = 3.412 × COP.

Likely exam questions

- State and explain Clausius and Kelvin-Planck statements; define COP, EER and TR. (short)
- Derive the COP of reversed Carnot cycle. Why is it not used in practice? (long)
- Explain the Bell-Coleman cycle with p-V and T-s diagrams; derive its COP. (long)
- Compare open and dense air refrigeration systems. (short)

- Explain refrigeration needs of aircraft; compare the types of air-cooling systems. (long)
- Numerical: air-refrigeration cycles with efficiencies; TR and power calculation.

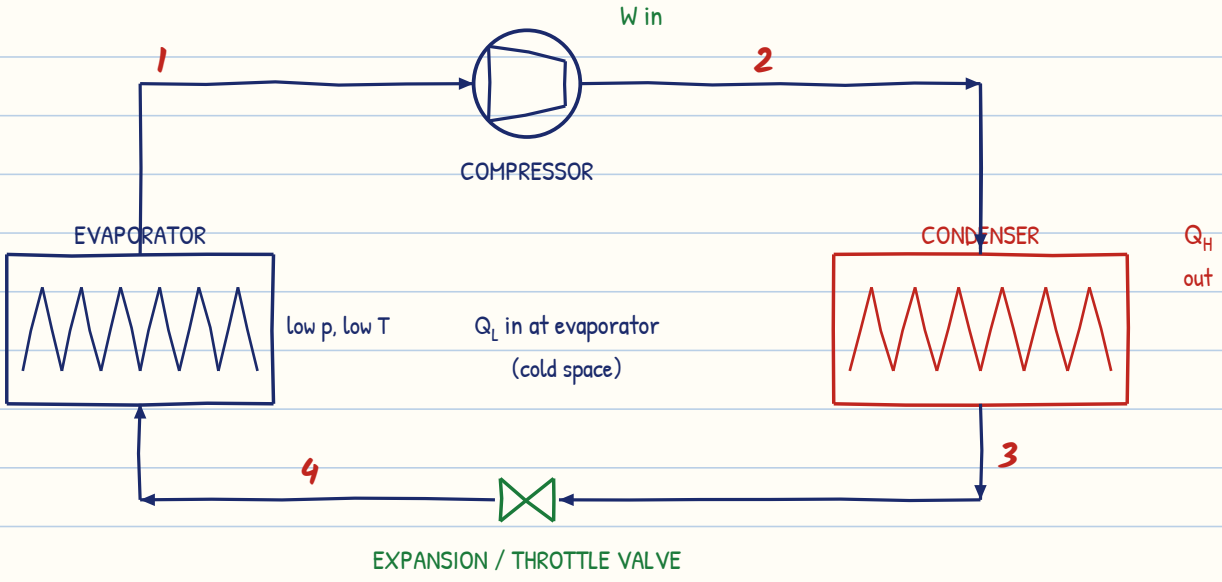
Syllabus: VCR system – basic cycle, working principle and components, COP, T-s and p-h charts, expander vs throttling, effect of sub-cooling and super-heating, cycle analysis, actual cycle, influence of parameters, p-h chart, numericals. Refrigerants – properties, classification, nomenclature, secondary refrigerants, lubricants, ozone depletion, global warming, newer refrigerants.

I. Vapour Compression Refrigeration (VCR) System

- Most widely used refrigeration system (domestic fridge, AC, cold storage, chillers).
- Working fluid is a **refrigerant** that changes phase: it absorbs the latent heat while **evaporating** at low temperature and rejects heat while **condensing** at high temperature.
- Mechanical energy (compressor) is the input; used for capacities from 50 W to many MW.

Essential components

- **Compressor** – raises pressure and temperature of vapour (reciprocating, rotary, scroll, screw, centrifugal). It circulates the refrigerant (the "heart" of the plant).
- **Condenser** – desuperheats, condenses and (sub-cools) the vapour; rejects Q_H to air or water.
- **Expansion device** – throttle valve / capillary tube / thermostatic expansion valve (TEV); reduces pressure from p_c to p_e and controls refrigerant flow.
- **Evaporator** – refrigerant boils at low pressure, absorbs Q_L from the cold space.
- Accessories: receiver, filter-drier, sight glass, accumulator, oil separator, pressure cut-outs.



Components in detail

Component	Types	Notes
Compressor	Reciprocating, rotary, scroll, screw, centrifugal	Recip. up to ~100 TR; centrifugal for > 200 TR chillers
Condenser	Air-cooled, water-cooled, evaporative	Air-cooled: small units; water: cooling tower; evaporative: ammonia plants
Evaporator	Dry-expansion, flooded, shell & tube, bare tube / finned coil	Flooded gives higher h, dry-expansion uses TEV
Expansion device	Capillary tube, TEV, automatic EV, float valve, electronic EV	Capillary: fixed orifice, domestic; TEV keeps constant superheat

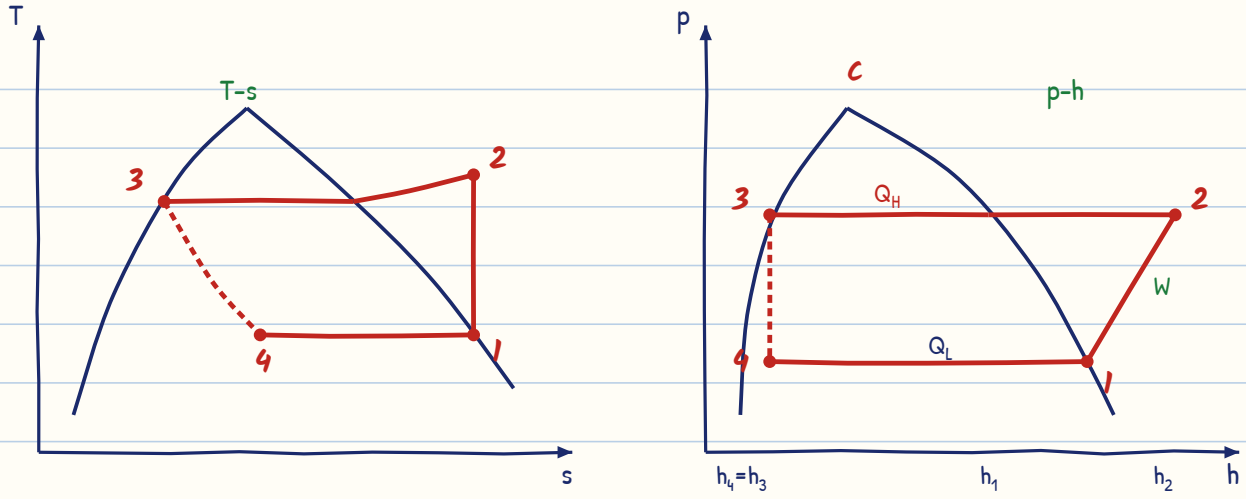
- **Receiver** stores liquid so that load variations are handled; **filter-drier** removes moisture and dirt; **accumulator** prevents liquid reaching compressor; **oil separator** returns oil to crankcase.
- **Safety controls:** HP/LP cut-out, oil pressure switch, thermostat, relief valve.

Hermetic, semi-hermetic and open compressors

- **Hermetic:** motor + compressor sealed in one shell (fridges, ACs) – no shaft seal leak, noisy-free, but not serviceable.
- **Semi-hermetic:** bolted shell, can be opened for service. **Open type:** motor outside, shaft seal, used for NH_3 (motor cannot be exposed to NH_3).

Working principle (simple saturated cycle)

- **1-2** Isentropic compression of dry saturated vapour from p_e to p_c. Vapour becomes superheated: T rises, work W input.
- **2-3** Constant pressure cooling and condensation in condenser (desuperheating 2-2', condensation 2'-3). Heat Q_H rejected. State 3 is saturated liquid.
- **3-4** Throttling (isenthalpic expansion) through the valve: p falls to p_e, temperature falls to T_e, part of the liquid flashes to vapour. $h_3 = h_4$.
- **4-1** Constant pressure, constant temperature evaporation. Refrigerant absorbs Q_L (latent heat) from the cold space.



Cycle analysis (per kg of refrigerant)

Refrigerating effect $q_L = h_1 - h_4 = h_1 - h_3$

Compressor work $w = h_2 - h_1$

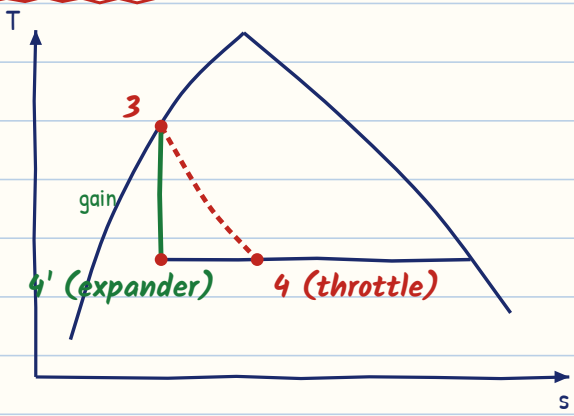
Heat rejected $q_H = h_2 - h_3 = q_L + w$

$$COP = (h_1 - h_4) / (h_2 - h_1) = (h_1 - h_3) / (h_2 - h_1)$$

- Mass flow rate $m = Q_L / (h_1 - h_4) = (TR \times 3.5167) / (h_1 - h_4)$ kg/s
- Compressor power $P = m (h_2 - h_1)$; Condenser load = $m (h_2 - h_3)$.
- Dryness fraction at 4: $x_4 = (h_4 - h_{f4}) / (h_{fg4})$ – the **flash gas** fraction, which gives no cooling.
- Volume flow at compressor inlet $V = m v_1$; swept volume = V/η_{vol}.

Note: Compare with the Carnot cycle: the VCR cycle replaces the wet compression with dry compression and the expander with a throttle – practical but a little less efficient.

2. Expander vs Throttling



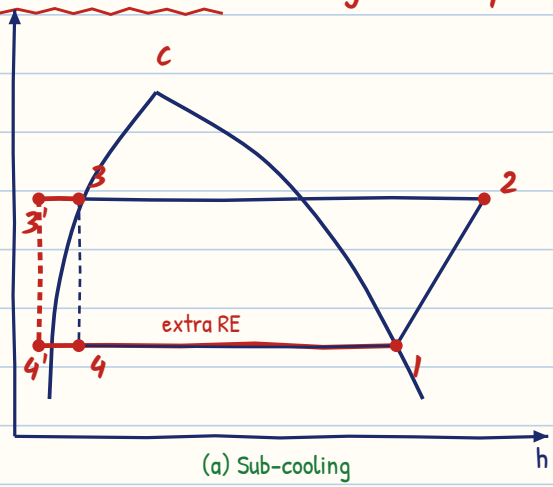
Throttle (3–4): h = const, irreversible, no work recovered, simple, cheap.

Expander (3–4'): s = const, work recovered, RE rises by ($h_4 - h_{4'}$), but complex & costly; liquid wet-expansion problems.

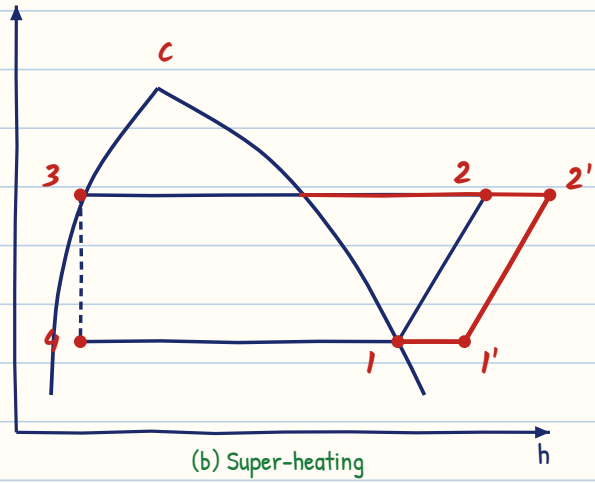
Gain is small (≈ 2–5 %) → throttle is used.

- With an expander the work recovered is ($h_3 - h_{4'}$) and COP improves, but liquid expansion makes work very small because the liquid is nearly incompressible; friction and cost cancel the gain.
- Throttle valve / capillary has no moving parts, is cheap and controls flow – therefore used in all practical VCR plants. (Expanders are used in CO_2 transcritical and some cryogenic plants where the gain is large.)

3. Effect of Sub-cooling and Super-heating



(a) Sub-cooling



(b) Super-heating

Sub-cooling (3 → 3')

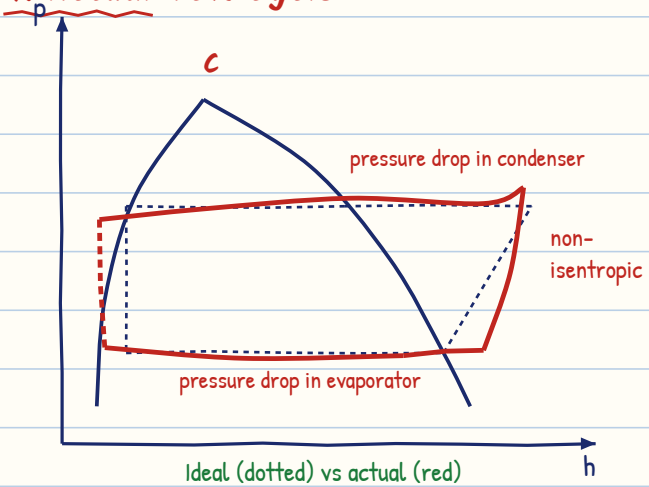
- Liquid cooled below saturation temperature at condenser pressure (by sub-cooler, liquid-suction heat exchanger, or extra length in condenser).
- Refrigerating effect increases ($h_1 - h_{4'}$), compressor work unchanged → **COP increases**. Also prevents flash gas before the valve.

Super-heating (1 → 1')

- Vapour leaves the evaporator slightly above saturation (5-10 K) – this protects the compressor from **liquid slugging** and helps TEV control.
- **Useful super-heating** (inside the cooled space) increases RE and COP slightly; **useless super-heating** (in suction line outside) increases work with little gain.
- Excessive super-heating raises discharge temperature and may break down oil.

Change	RE	Work	COP
Sub-cooling	Increases	Same	Increases
Super-heating (useful)	Increases	Increases	Increases / almost same
Super-heating (useless)	Same	Increases	Decreases
Higher evaporator temperature	Increases	Decreases	Increases
Higher condenser temperature	Decreases	Increases	Decreases

4. Actual VCR Cycle



Actual cycle differs from ideal:
- suction superheat (1-1')
- pressure drops in lines, valves
- compression not isentropic
- sub-cooling in condenser
- heat gain in suction line
- valve losses, leakage

- Real compression has friction and heat transfer → entropy increases, $\eta_{isen} = (h_{2s} - h_1)/(h_2 - h_1) \approx 0.7\text{-}0.85$.
- **Volumetric efficiency** $\eta_{vol} < 1$ due to clearance, valve leakage, wire drawing and heat transfer.
- Superheating at inlet, sub-cooling at outlet of condenser, pressure drop everywhere: the evaporator must be at a lower p and the condenser at a higher p than ideal cycle → more work.

- Real COP is only 50-65 % of the ideal cycle value.

Combined sub-cooling + super-heating cycle

Let h_1 = enthalpy after super-heating, h_2 = after compression from 1, h_3 = enthalpy of sub-cooled liquid = h_4.

$$COP = (h_1 - h_3) / (h_2 - h_1) \text{ ; isentropic efficiency } \eta = (h_{2s} - h_1) / (h_2 - h_1)$$

- Sub-cooled liquid enthalpy can be approximated by h_f at its own temperature: $h = h_f(T_{sub})$.
- Super-heated vapour enthalpy: $h_1 \approx h_g + c_{pv} (T_{sup} - T_{sat})$.

Liquid-suction heat exchanger (LSHX)

- A small counterflow exchanger that sub-cools liquid from condenser using the cold suction vapour: heat balance $h_3 - h_{3'} = h_{1'} - h_1$.
- Benefits: avoids flash gas, ensures dry suction; effective with R134a, R12; poor for NH_3 (high discharge temp).

5. Influence of Various Parameters on System Performance

Evaporator pressure / temperature (T_e)

- Raising T_e: RE ↑, specific volume at suction ↓ (so refrigerant flow rises), compressor work per kg ↑ only slightly → **COP ↑ (about 2-4 % per K)**.
- Lowering T_e reduces capacity sharply, needs a bigger compressor, and may lead to frosting.

Condenser pressure / temperature (T_c)

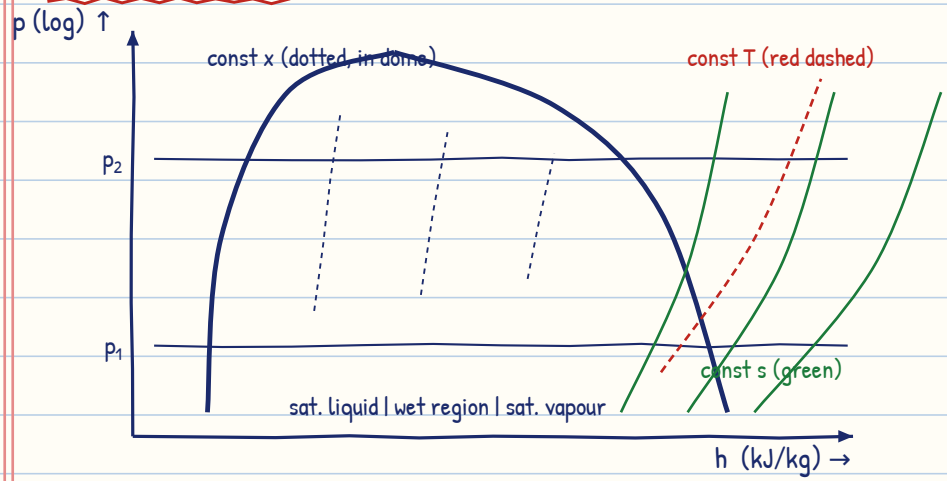
- Raising T_c: work ↑, RE ↓ (due to more flash gas), discharge temp ↑ → **COP ↓, capacity ↓**.
- Keep condenser clean; use cooling water or large air flow; avoid non-condensable gases.

Others

- **Sub-cooling and super-heating** as above. **Compressor speed / clearance:** affects η_{vol}.
- **Type of refrigerant:** latent heat, specific volume decide the flow rate and size.
- **Moisture and oil in system:** choke the valve, reduce heat transfer.

> **Note:** For every 1 K rise in condensing temperature the power input increases by roughly 2-3 %.

6. Construction and Use of p-h Chart



Lines on p-h chart:

- Saturation dome (x = 0, x = 1)
- Isobars – horizontal
- Isenthalpic – vertical
- Isotherms – vertical in liquid, horizontal in dome, steep in vapour
- Isentropes – rising to right
- Constant volume & quality lines

- The p-h (Mollier) chart plots **pressure (log scale, y)** against **enthalpy (x)**, each refrigerant has its own chart.
- It gives every enthalpy needed (h_1, h_2, h_3, h_4) directly as **horizontal lengths** – hence the cycle is easy to calculate.
- **Construction (steps):** (i) draw the saturated liquid and vapour curves meeting at the critical point; (ii) draw isobars (horizontal) at various pressures; (iii) join equal enthalpy points to draw vertical isenthalpic lines; (iv) draw isotherms: vertical in sub-cooled liquid, horizontal in wet region, falling steeply in superheated zone; (v) add isentropic lines (s = const), constant volume and dryness fraction lines.

Using the chart for a simple saturated cycle

- Locate evaporator pressure line; mark 1 on the saturated vapour line.
- From 1 follow the isentrope up to p_c to get 2; read h_2.
- From 2 go horizontally left to the saturated liquid line to get 3; drop vertically to p_e to get 4 ($h_4 = h_3$).
- Read h_1, h_2, h_3 and use the COP formula.

7. Numerical Problems

Problem 1

A refrigerator using R-134a works between −10 °C (evaporator) and 40 °C (condenser) on a standard saturated cycle. From tables: h_1 = 392.3, h_2 = 424.0, $h_3 = h_4$ = 256.4 kJ/kg (values approximate). Find RE, work, COP, and the mass flow & power for 5 TR.

RE = 392.3 - 256.4 = **135.9 kJ/kg** ; w = 424.0 - 392.3 = **31.7 kJ/kg**

COP = 135.9/31.7 = **4.29**

m = 5 × 3.5167 / 135.9 = **0.129 kg/s** ; P = 0.129 × 31.7 = **4.10 kW**

Condenser heat = m (h_2 - h_3) = 0.129 × 167.6 = 21.7 kW (check: 17.58 + 4.10)

Problem 2 (effect of sub-cooling)

In problem 1 the liquid is sub-cooled to 35 °C (h = 249.0 kJ/kg). Find new COP. (same compressor work approx.)

RE = 392.3 - 249.0 = 143.3 kJ/kg ; COP = 143.3/31.7 = **4.52** → increase of 5.4 %.

Problem 3 (given enthalpies)

A VCR plant of 10 TR has h_1 = 380, h_2 = 430, h_3 = 240 kJ/kg. Find COP, mass flow, power, condenser load.

RE = 140 ; w = 50 ; COP = 140/50 = **2.8**

m = 35.167/140 = **0.251 kg/s** ; P = 0.251 × 50 = **12.56 kW** ; Q_H = 0.251×190 = **47.7 kW**

Problem 4 (volumetric flow)

In problem 3, suction specific volume is 0.07 m^3/kg, η_{vol} = 0.8. Swept volume = m v_1/η_{vol} = 0.251 × 0.07/0.8 = **0.022 m^3/s** = 79 m^3/h.

Problem 5 (relative COP)

Carnot COP for −10 °C and 40 °C = 263/50 = 5.26. Relative COP for problem 1 = 4.29/5.26 = **0.82**.

Problem 6 (ammonia plant, data given)

A 20 TR ammonia plant has h_1 = 1450, h_2 = 1670, h_3 = h_4 = 340 kJ/kg. Find COP, mass flow, power.

RE = 1450 - 340 = 1110 kJ/kg ; w = 220 kJ/kg ; COP = 1110/220 = **5.05**

$m = 20 \times 3.5167/1110 = \mathbf{0.0634\ kg/s}$; $P = 0.0634 \times 220 = \mathbf{13.9\ kW}$

Problem 7 (super-heating + sub-cooling)

An 8 TR plant: $h_1 = 395$, $h_2 = 432$ (after super-heat), $h_3 = h_4 = 245$ kJ/kg (sub-cooled).

$RE = 150$; $w = 37$; $COP = 150/37 = \mathbf{4.05}$; $m = 28.13/150 = \mathbf{0.1876\ kg/s}$; $P = 0.1876 \times 37 = \mathbf{6.94\ kW}$

Problem 8 (isentropic efficiency of compressor)

In problem 1, the compressor isentropic efficiency is 80 %. Actual $h_2 = 392.3 + (424.0 - 392.3)/0.8 = 431.9$ kJ/kg.

$w_{actual} = 39.6$ kJ/kg ; $COP_{actual} = 135.9/39.6 = \mathbf{3.43}$ (instead of 4.29)

Problem 9 (power per TR)

For problem 1: power per TR = $4.10/5 = \mathbf{0.82\ kW/TR}$, EER = $3.412 \times 4.29 = 14.6$.

8. Refrigerants

A refrigerant is the working fluid that absorbs heat by evaporating and rejects heat by condensing.

Desirable properties

- **Thermodynamic:** high latent heat; low boiling point at atmospheric pressure; evaporator pressure above atmosphere (no air leak-in); moderate condenser pressure; low critical-point issues (critical T well above T_c); low freezing point; low specific volume of vapour; high COP; low discharge temperature.
- **Physical / chemical:** chemically stable, non-corrosive, non-reactive with oil and metals; good heat transfer (high conductivity), low viscosity; miscible with oil as required.
- **Safety:** non-toxic, non-flammable, non-explosive; easy leak detection; low cost and easily available.
- **Environment:** zero ODP and low GWP.

Classification

Group	Examples	Remarks
Halocarbons – CFC	R11, R12, R113	ODP high, phased out
HCFC	R22, R123, R141b	ODP small, being phased out
HFC	R134a, R32, R125, R152a	ODP 0, GWP moderate
HFO	R1234yf, R1234ze	GWP < 1, mildly flammable
Inorganic	R717 (NH_3), R718 (water), R744 (CO_2), R729 (air)	Natural refrigerants
Hydrocarbons	R290 (propane), R600a (isobutane), R1270	Flammable, zero ODP, tiny GWP
Azeotropes	R500, R502, R507A	Behave as pure fluid (R-5xx)
Zeotropes (blends)	R404A, R407C, R410A	Temp glide (R-4xx)

Nomenclature (ASHRAE numbering)

- For R-**XYZ** (hydrocarbon derivatives): X = C - 1 (omit if zero), Y = H + 1, Z = number of F atoms. Remaining valence is Cl.

R12 = CCl_2F_2 (C-1 = 0, H+1 = 1, F = 2)

R22 = $CHClF_2$ (0, 2, 2) R134a = $C_2H_2F_4$ (1, 3, 4; 'a' = isomer)

R32 = CH_2F_2 (0, 3, 2) R11 = CCl_3F (0, 1, 1)

- Inorganic: R-**7XX** with XX = molecular mass: NH_3 = R717, water = R718, CO_2 = R744.
- Blends: R-4XX (zeotropic), R-5XX (azeotropic). R-4xx with letter: composition, e.g. R410A = R32/R125 (50/50).
- Isobutane = R600a, propane = R290.

NH_3 versus R134a (comparison)

Property	R717 (NH_3)	R134a
Latent heat at −15 °C	~1315 kJ/kg	~210 kJ/kg
Boiling point at 1 atm	−33.3 °C	−26.3 °C
Toxicity / flammability	Toxic, B2 (mild flammable)	Non-toxic, A1
Compatibility	Not with copper	Compatible
Application	Industrial, cold storage	Domestic, car AC
ODP / GWP	0 / 0	0 / 1430
Leak detection	Smell, sulphur candle	Halide torch, electronic

Mixtures

- **Azeotrope:** boils at a constant temperature like a pure fluid (R502 = R22 + R115; R507A). **Zeotrope:** composition changes during phase change, so there is temperature **glide** (R407C glide ≈ 5 K).
- R410A is a near-azeotrope with small glide, runs at ≈ 60 % higher pressure than R22.

Secondary refrigerants (brines)

- Heat-carrying fluid cooled by the primary refrigerant, then circulated to the load (indirect system) – used in large plants, ice rinks, breweries, to reduce primary charge.
- Examples: water (above 0 °C), **NaCl brine** (eutectic 23.1 %, −21.1 °C), **$CaCl_2$ brine** (29.9 %, −55 °C), **ethylene / propylene glycol** (non-toxic, for food), methanol, CO_2.
- Needs: low freezing point, high specific heat, low viscosity, non-corrosive (inhibitors), cheap.

Lubricants

- Compressor oil lubricates, seals, cools. Properties: low pour point, correct viscosity, high flash point, chemical stability, miscibility with refrigerant, low wax formation (floc point).
- **Mineral oil** (CFC, HCFC, NH_3), **alkylbenzene**, **POE** (polyol-ester, for HFC), **PAG** (auto AC with R134a), **PAO**. HFCs are not miscible with mineral oil – use POE.

Ozone depletion and global warming

- Chlorine from CFC/HCFC in the stratosphere destroys ozone: $Cl + O_3 \rightarrow ClO + O_2$. **ODP** is relative to R11 = 1. R12 = 1, R22 = 0.055, R134a = 0.
- **GWP** relative to CO_2 = 1 (100 yr): R12 ≈ 10 900, R22 ≈ 1 810, R134a ≈ 1 430, R410A ≈ 2 090, R32 ≈ 675, R1234yf ≈ 4, R290 ≈ 3, NH_3 = 0.

- **Montreal Protocol (1987)** – phase-out of CFCs and HCFCs. **Kigali Amendment (2016)** – phase-down of HFCs. **Kyoto Protocol** – greenhouse gases.

Refrigerant	ODP	GWP	Notes
R12	1.0	~10900	banned
R22	0.055	~1810	phasing out
R134a	0	~1430	fridge, car AC
R410A	0	~2090	split AC
R32	0	~675	new split AC
R290	0	~3	flammable
R717 (NH_3)	0	0	toxic, industrial

Newer refrigerants

- **R32** (lower GWP, A2L), **R1234yf** (car AC), **R1234ze** (chillers), **R290 / R600a** (domestic), **R744 (CO_2)** transcritical systems, **R454B / R513A** (low-GWP blends replacing R410A / R134a), natural refrigerants NH_3.
- Trend: ZERO ODP, low GWP, mildly flammable (A2L) with safety standards.

Safety classification of refrigerants (ASHRAE 34)

- Letter = toxicity: **A** = lower toxicity, **B** = higher toxicity. Number = flammability: **1** = no flame propagation, **2L** = mildly flammable, **2** = flammable, **3** = highly flammable.

Class	Example
A1	R134a, R410A, R22, CO_2
A2L	R32, R1234yf, R1234ze
A3	R290, R600a
B2L	R717 (ammonia)

Leak detection and charging

- Halide torch (flame turns green), soap bubble test, electronic detector, UV dye; ammonia – sulphur candle (white smoke) or litmus paper.
- Always **evacuate** the system with a vacuum pump (below 500 microns) before charging – removes air and moisture.
- Charge by weight; check superheat (TEV) or sub-cooling (capillary systems).

Example: effect of T_e on COP

For R134a at T_c = 40 °C: COP(T_e = −10 °C) ≈ 4.3, COP(T_e = −20 °C) ≈ 3.2, COP(T_e = 0 °C) ≈ 6.0. So raising evaporator temperature by 10 K increases COP by about 40 %. This is why chillers are run at the highest possible evaporator temperature.

9. Quick Revision and Exam Questions

- COP = $(h_1 - h_4)/(h_2 - h_1)$; m = TR×3.5167/RE; $h_3 = h_4$.
- Sub-cooling ↑ COP; ↑T_e ↑ COP; ↑T_c ↓ COP.
- R-number rule (C-1)(H+1)(F); ODP of R11 = 1; GWP of CO_2 = 1.

Likely questions

- Draw VCR cycle on T-s and p-h charts; derive COP; list components with functions. (long)
- Compare expander and throttle valve. (short)
- Discuss effect of sub-cooling, super-heating, evaporator and condenser temperatures on COP. (long)
- Explain actual VCR cycle and its deviations. (long)
- Properties of a good refrigerant; nomenclature of R12, R22, R134a, R717; secondary refrigerants. (long)
- ODP, GWP, Montreal and Kigali, newer refrigerants. (short)
- Numerical on VCR with given enthalpies: COP, mass flow, power.

Syllabus: Vapour absorption (VAR) system – description and working of NH_3-water, LiBr-water (two shell and four shell), maximum COP, three-fluid absorption system. Steam jet refrigeration – principle, components, estimation of motive steam. Thermo-electric refrigerator, vortex (Hilsch) tube.

I. Vapour Absorption Refrigeration (VAR) System

- Heat energy (steam, hot water, waste heat, solar, gas flame) drives the system instead of mechanical work. The compressor of VCR is replaced by the **absorber + pump + generator**.
- Two working substances: **refrigerant** (evaporates) and **absorbent** (absorbs the refrigerant vapour). Pair examples: NH_3 (refrigerant) – water (absorbent); water (refrigerant) – LiBr (absorbent).

Principle

- Refrigerant vapour at low pressure is **absorbed** in the weak solution (exothermic, cooled by water). Absorbing needs far less energy than compressing – a liquid can be pumped with 1-2 % of the work needed for vapour.
- Strong solution is pumped to high pressure and **heated** in the generator to boil off refrigerant vapour. This vapour goes to the condenser, throttle valve, evaporator – the same as in VCR.

> Compressor in VCR ≈ Absorber + Pump + Generator (+ heat exchanger) in VAR

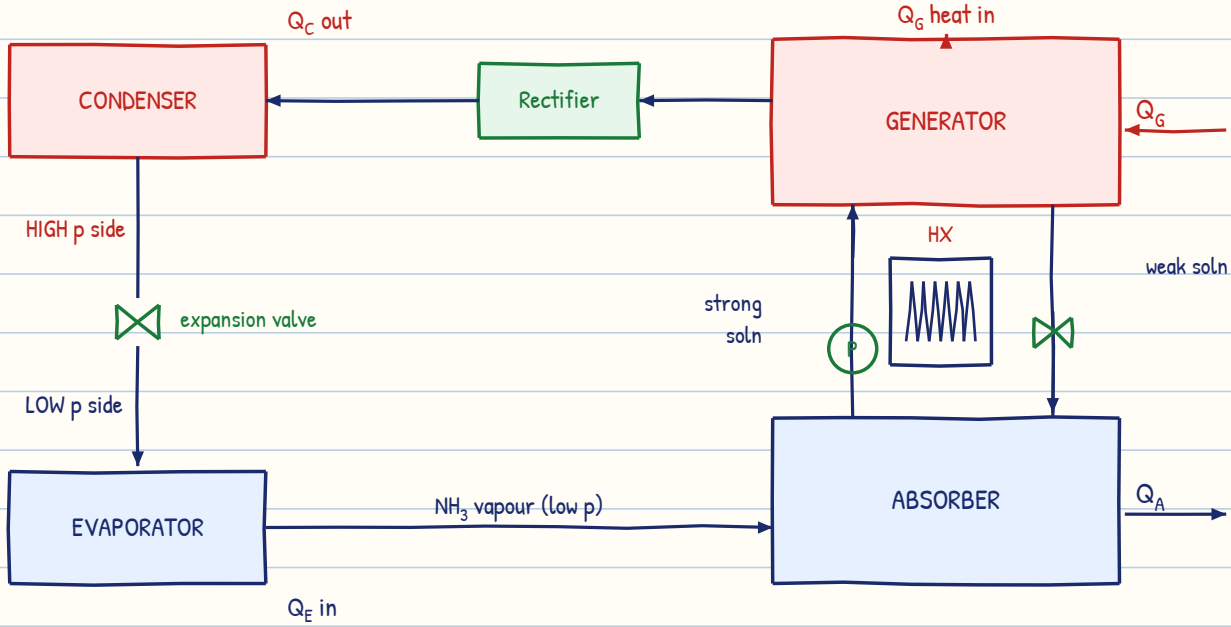
Advantages / disadvantages over VCR

- **Pros:** uses low-grade heat / waste heat; no moving parts except small pump → quiet, less vibration, reliable; capacity control easy (from 100 % to 10 %); no ODP / GWP refrigerant.
- **Cons:** low COP (0.5 - 1.2) so larger heat rejection; bulky and costly; NH_3 is toxic; LiBr crystallises; needs cooling tower.

Property	VCR	VAR
Energy input	Work (electricity)	Heat (steam, gas, waste)
Moving parts	Compressor (many)	Only small pump
COP	3 - 5	0.5 - 1.2

Property	VCR	VAR
Noise / vibration	Higher	Low
Part-load	Poor	Good
Size & cost	Smaller	Bigger, costlier
Best use	Everywhere	Where cheap heat is available

2. NH_3 – Water Absorption System (Description and working)



Components and functions

- **Evaporator:** liquid NH_3 at low p evaporates, absorbing Q_E from the cold space.
- **Absorber:** NH_3 vapour is absorbed by cold weak solution (water poor in NH_3) → strong solution. Heat of absorption Q_A is removed by cooling water. Low T of absorber helps absorption.
- **Pump:** raises pressure of strong solution from p_e to p_c (small work).
- **Heat exchanger:** strong solution (cold) picks up heat from the weak solution (hot) → saves generator heat and cools the weak solution before absorber.
- **Generator:** strong solution heated (steam / gas) and NH_3 vapour driven out. Water also vaporises a little as the boiling points are close.
- **Analyser and Rectifier (dephlegmator):** remove water vapour from NH_3 vapour (water vapour would freeze in the evaporator and choke the valve). Water condenses and drains

back to generator. NH_3 reaching condenser is about 99.8 % pure.

- **Condenser:** NH_3 vapour condensed by cooling water. **Expansion valve:** throttles the liquid to p_e.

Working in sequence

- Pure NH_3 vapour from generator → rectifier → condenser (Q_C rejected) → liquid → expansion valve → evaporator (Q_E absorbed) → vapour → absorber → strong solution → pump → heat exchanger → generator. The weak solution flows back through the heat exchanger and pressure-reducing valve to absorber.

> **Note:** In NH_3-water, the refrigerant (NH_3) and the absorbent (water) are both volatile → rectifier is necessary. In LiBr-water, LiBr is non-volatile → no rectifier is needed.

Practical improvements

- Heat exchanger between strong and weak solution; sub-cooler (liquid-vapour HX) to raise RE; rectifier with analyser to purify; proper cooling of absorber.

3. Maximum COP of an Absorption System

Model the system as a reversible **heat engine** (between T_G and T_0) driving a reversible **refrigerator** (between T_E and T_0). Three temperatures: generator T_G, surroundings (condenser & absorber) T_0, evaporator T_E.

$$COP_{actual} = Q_E / Q_G \text{ (pump work neglected)}$$

$$\text{Engine efficiency } \eta_{max} = (T_G - T_0) / T_G \text{ ; Refrigerator COP} = T_E / (T_0 - T_E)$$

$$\mathbf{COP_{max} = (T_G - T_0) / T_G \times T_E / (T_0 - T_E)}$$

- Maximum COP rises with higher T_G, higher T_E, lower T_0. Real COP is 60-80 % of this.

Example 1

T_G = 100 °C (373 K), T_0 = 30 °C (303 K), T_E = −10 °C (263 K):

$$COP_{max} = (70/373) \times (263/40) = 0.1877 \times 6.575 = \mathbf{1.23}$$

Example 2

A NH_3-water plant (T_G = 90 °C, T_0 = 35 °C, T_E = −5 °C): COP_{max} = (55/363) × (268/40) = 0.1515 × 6.7 = **1.015**. Heat needed for 10 TR at actual COP 0.6: Q_G = 35.17/0.6 = 58.6 kW.

4. LiBr – Water Absorption System

- Refrigerant = **water**, absorbent = **lithium bromide** (a salt, non-volatile). Evaporator works at ≈ 6 mm Hg abs (≈ 4 °C), so the plant is under high vacuum. Used for **air-conditioning and chilled water** (T_e > 0 °C only, as water freezes).
- Absorbent properties: strong affinity for water vapour, stable, non-toxic, no rectifier needed. Problems: **crystallisation** at high concentration (> 65 %) and low temperature; corrosive – needs inhibitors; vacuum maintained by purge unit.

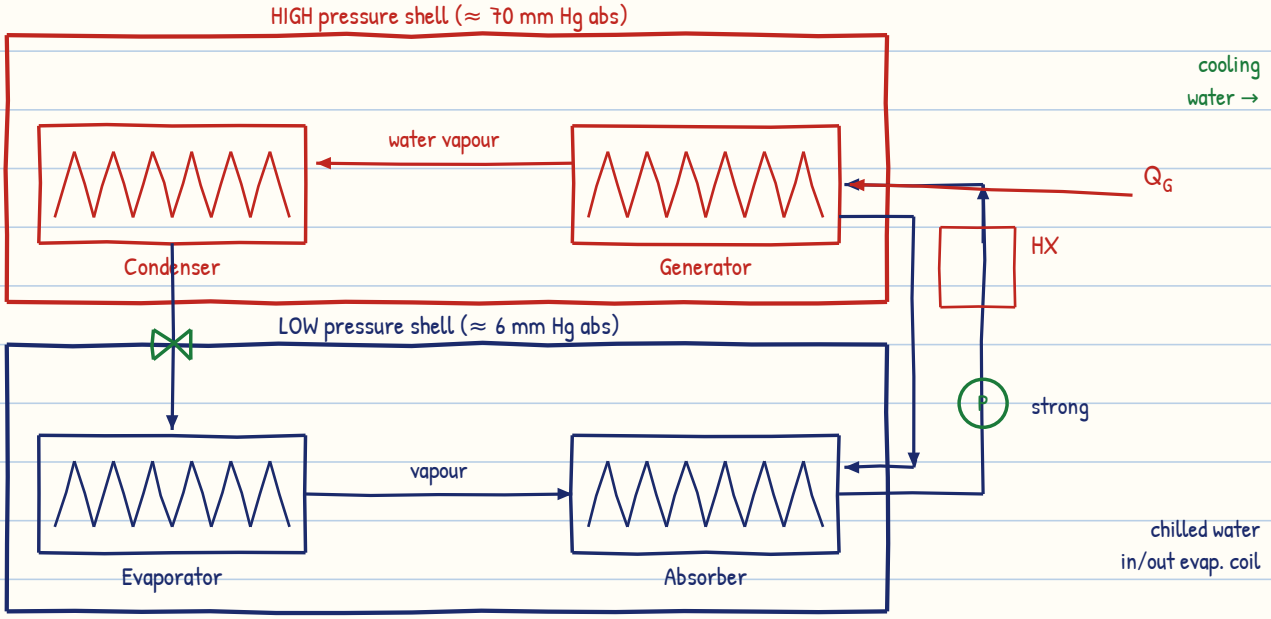
Solution concentration and Dühring chart

- The state of LiBr-water solution is plotted on a **Dühring chart** (solution temperature vs refrigerant saturation temperature, lines of constant concentration) – it shows how the same vapour pressure needs a higher temperature as the concentration rises.
- Typical cycle: absorber outlet ≈ 58 % LiBr at 35-40 °C (dilute); generator outlet ≈ 63-64 % at 90-100 °C (strong). The difference in concentration is called the **degassing range**; limits crystallisation.
- **Crystallisation control:** keep the concentration below the crystallisation line; dilution cycle at shutdown; avoid sudden drop in cooling water temperature.

Double-effect LiBr-water

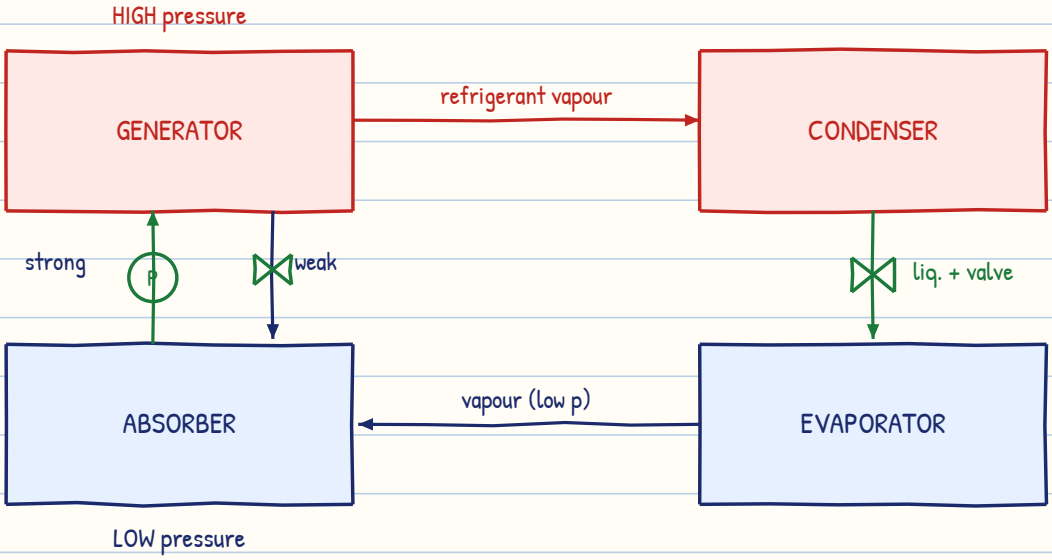
- Two generators: the high-temperature generator (≈ 150 °C, driven by steam at 8 bar or gas flame) boils vapour that is then used as the **heating medium of the low-temperature generator**; the heat is used twice → COP ≈ 1.0-1.3.

Two-shell system



- ***Upper (high-pressure) shell:*** *generator + condenser.* ***Lower (low-pressure) shell:*** *evaporator + absorber.*
- *Dilute solution from absorber is pumped through the heat exchanger to the generator; steam / hot water boils water vapour out of it. Vapour condenses (cooling water), liquid water is throttled (U-tube / orifice) to evaporator and sprayed on chilled-water coil – evaporates and cools the chilled water (e.g. 12 → 7 °C).*
- *Concentrated solution returns through HX to the absorber and is sprayed over the cooling coil; it absorbs the water vapour and is diluted. Cooling water goes first to absorber, then condenser.*
- *Most common type for small-medium sizes (single-effect, COP ≈ 0.7).*

Four-shell system



- Generator, condenser, evaporator and absorber are placed in **four separate shells** connected by pipes. It allows flexible layout, easy maintenance and use for very large capacity or **double-effect** chillers (two generators, COP ≈ 1.2).
- Used where height is limited or when higher temperature heat source is available (direct-fired). Costlier due to more shells and piping.

Feature	Two shell	Four shell
Shells	2 (HP & LP)	4 (each component)
Size & cost	Compact, cheaper	Large, costlier
Maintenance	Harder (combined)	Easy
Use	Small & medium	Large / double-effect

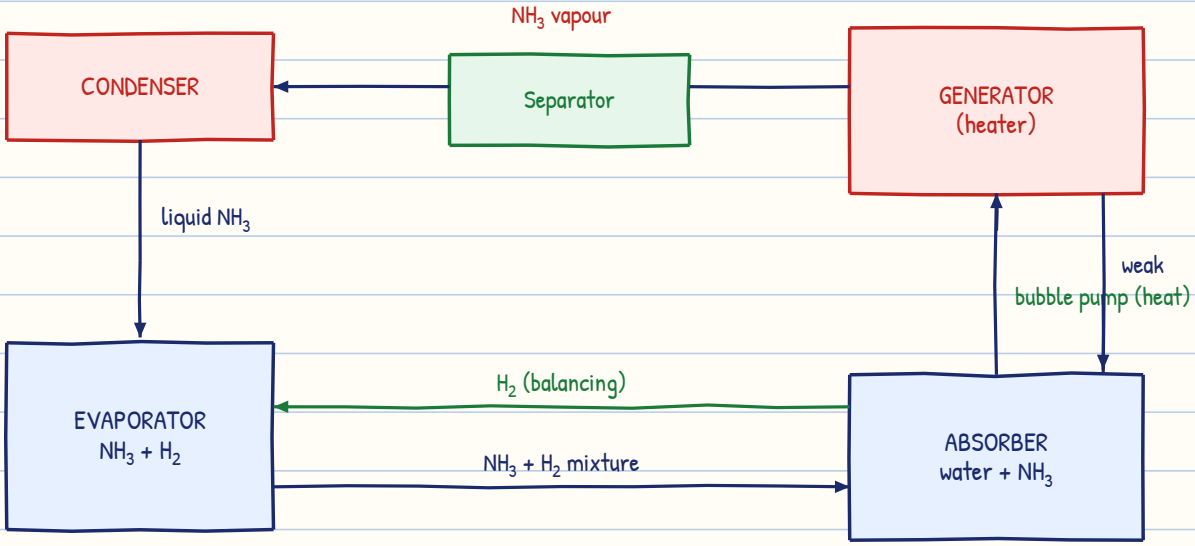
NH_3-water vs LiBr-water

NH_3-water	LiBr-water
Refrigerant volatile → rectifier	Absorbent non-volatile → no rectifier
Can reach below 0 °C	Only above 0 °C (water refrigerant)
Works at above atm pressure	Works under high vacuum
Toxic (NH_3)	Non-toxic
Cools industrial, cold storage	Used for air conditioning
COP ≈ 0.5	COP ≈ 0.7 - 1.2

> **Note:** Water has a very high latent heat (2500 kJ/kg), hence LiBr-water has better COP than NH_3-water.

5. Three-Fluid Absorption System (Platen-Munters)

- Invented by Platen and Munters (Sweden). Three fluids: **NH_3** (refrigerant), **water** (absorbent) and **hydrogen** (inert gas that maintains the **total pressure equal** throughout the plant).

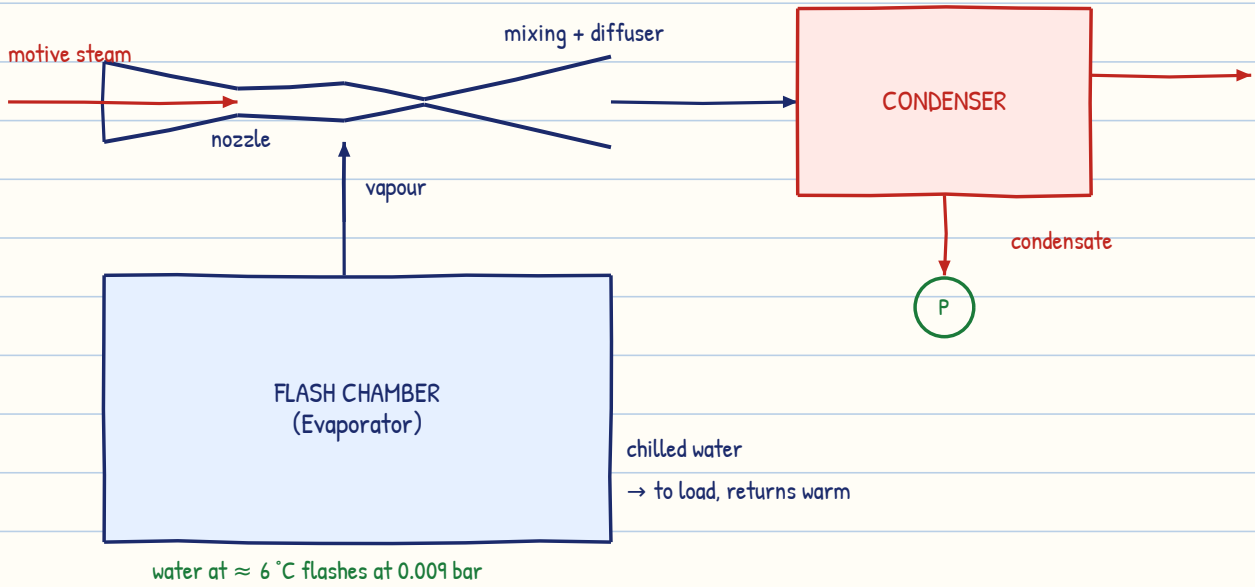


Total pressure same everywhere – no pump, no valve

- Total pressure is the same everywhere (≈ 15 bar), so **no pump and no expansion valve** are needed – circulation is by density difference (**bubble pump / thermosyphon**) and gravity. NH_3 evaporates in the evaporator because its **partial pressure** is lowered by H_2: $p_{total} = p_{NH3} + p_{H2}$.
- Heat from electric heater / kerosene flame lifts weak solution by bubble pump; NH_3 vapour condenses; liquid NH_3 flows to evaporator where it diffuses into H_2 and absorbs heat; the NH_3-H_2 mixture goes to absorber where water absorbs NH_3, and H_2 returns.
- **Uses:** silent domestic refrigerators, hotel mini-bars, caravans (LPG) – no moving parts. COP is low (≈ 0.2-0.3).

6. Steam Jet Refrigeration System

- Water is used as the refrigerant! Water is evaporated at a very low pressure (≈ 0.009 bar at 6 °C) in a **flash chamber**; latent heat comes from the remaining water, which is **chilled**. High pressure steam acts through an **ejector** to maintain the vacuum and to compress the vapour.



Basic components

- **Flash chamber (evaporator):** warm return water flashes at low p; unflashed water is chilled to 4-10 °C.
- **Ejector:** convergent-divergent **nozzle** (steam expands to p_e, high velocity ≈ 1000 m/s), **suction / mixing chamber** (entrains vapour from flash chamber) and **diffuser** (velocity → pressure).
- **Condenser:** condenses motive steam + flash vapour (surface or barometric type). **Air ejector** removes air; **hot well & pumps** return condensate to boiler and make-up water to flash chamber.

Working

- Steam at 8-10 bar → nozzle (p_e ≈ 0.01 bar) → entrains flash vapour → diffuser raises p to condenser (≈ 0.05 bar) → condensed. Chilled water is circulated to the load (air handler).

Estimation of motive steam

- Fraction flashed: $x = c_{pw} (T_{in} - T_{out}) / h_{fg}$. Flash vapour $m_v = Q_L / h_{fg}$ (when cooling duty Q_L is given).

- Energy balance on nozzle / diffuser: kinetic energy of steam jet is shared with the entrained vapour:

$$m_s\,(h_s - h_n)\,\eta_n\,\eta_d = m_v\,(h_d - h_v) \;[\text{approx.}]$$

Entrainment ratio $ER = m_v/m_s$; motive steam $m_s = m_v/ER$ ($ER \approx 0.3 - 0.6$)

Example

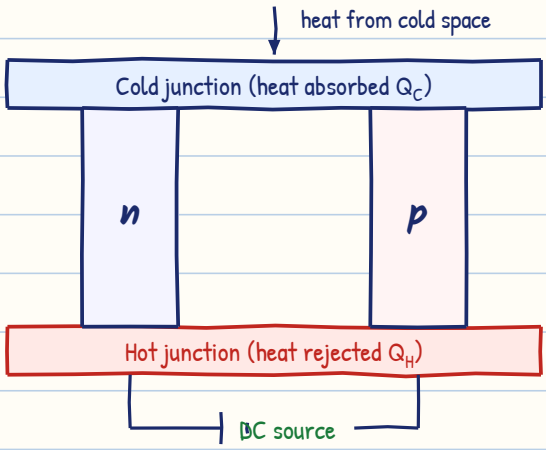
Cooling load 10 TR at 6 °C (h_{fg} = 2487 kJ/kg): $m_v = 35.17/2487 = 0.01414$ kg/s. If ER = 0.5, $m_s = 0.0283$ kg/s = **102 kg/h** ≈ 10 kg/h per TR.

Merits and demerits

- Merits: no moving parts, no harmful refrigerant, cheap, uses waste steam; handles large loads.
- Demerits: low COP (≈ 0.3), high steam consumption, can't go below 0 °C, ejector designed for one load, need cooling water.

7. Thermo-electric Refrigerator

- Based on the **Peltier effect** (reverse of Seebeck): when current passes through a circuit of two dissimilar conductors (n-type and p-type semiconductor), one junction cools and the other heats.



Peltier effect: when DC passes through a junction of two dissimilar semiconductors (Bi_2Te_3), one junction cools, other heats.

$Q_C = I\,T_C - ½\,I^2R - K\,\Delta T$

$W = I\,\Delta T + I^2R$

- Electrons in n-type material carry heat from cold junction to hot junction; semiconductors (Bi_2Te_3) have high Seebeck coefficient , low thermal conductivity K and low resistance R (figure of merit $Z = {}^2/(K\,R)$).

- Heat sink (fins / fan) on hot side. Many couples in series (thermopile) give higher capacity.

$$Q_C = \alpha I T_C - \tfrac{1}{2} I^2 R - K (T_H - T_C) \; ; \; W = \alpha I (T_H - T_C) + I^2 R \; ; \; COP = Q_C / W$$

Example

Module with $\alpha = 0.05$ V/K, $R = 2\ \Omega$, $K = 0.5$ W/K, $T_C = 273$ K, $T_H = 303$ K, $I = 2$ A:

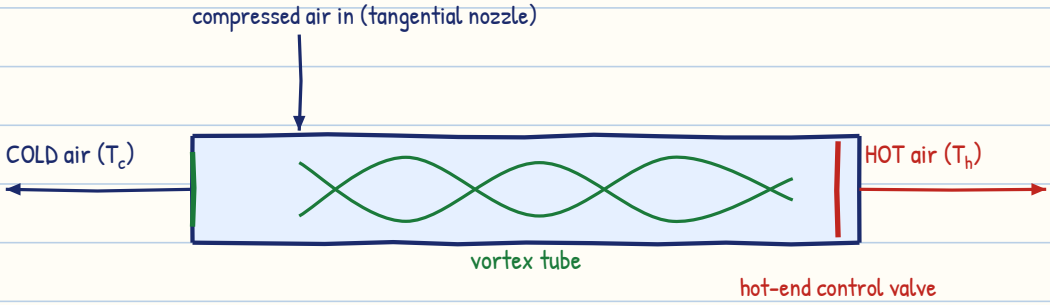
$$Q_C = 0.05 \times 2 \times 273 - 0.5 \times 4 \times 2 - 0.5 \times 30 = 27.3 - 4 - 15 = \mathbf{8.3\ W}$$

$$W = 0.05 \times 2 \times 30 + 4 \times 2 = 3 + 8 = 11\ W \; ; \; COP = 8.3/11 = \mathbf{0.75}$$

- **Advantages:** no moving parts, silent, compact, no refrigerant, reversible (heating also), precise control. **Disadvantages:** low COP, high cost, needs DC, small capacity. **Uses:** portable coolers, car mini-fridge, electronic chip cooling, medical, space, water coolers.

8. Vortex Tube (Hilsch Tube)

- Invented by Ranque, developed by Hilsch. No moving parts: compressed air (5–7 bar) is blown **tangentially** into a tube, producing a rotating vortex. The outer layers become **hot** and leave through the hot-end valve; the inner core becomes **cold** and leaves through the orifice at the other end.



$$\mu_c T_c + (1 - \mu_c) T_h = T_i$$

μ_c = cold mass fraction = m_c / m_i

- **Why separation occurs:** (i) the free vortex rotates faster at the centre, so the core loses energy by shear to outer layers (angular momentum, friction); (ii) expansion cools the core, work done on the outer layers heats them.

- Cold mass fraction $_c = m_c/m_i$. Cold-end orifice gives lowest T when $_c \approx 0.3$; maximum cooling effect ($_c \cdot \Delta T_c$) at $_c \approx 0.6$-0.7.

> Energy balance: $_c T_c + (1 - _c) T_h = T_i$

Example

Inlet air 300 K, $_c = 0.4$, $T_c = 270$ K → $T_h = (300 - 0.4\times270) / 0.6 =$ **320 K**. Refrigeration effect = $m_c c_p (T_i - T_c) = 0.4\ m \times 1.005 \times 30 = 12\ m$ kJ/s.

- **Advantages:** simple, cheap, no moving parts, maintenance free, instant cooling, compact. **Disadvantages:** low efficiency (COP ≈ 0.1), noisy, needs compressed air. **Uses:** spot cooling of machine tools, soldering, electronic cabinets, CNC machining, cooling of suits.

Problem 1 (heat balance of absorption plant)

A VAR plant gives 50 kW cooling with actual COP 0.7. Find generator heat and heat rejected.

> $Q_G = 50/0.7 =$ **71.4 kW** ; heat rejected (condenser + absorber) = $Q_G + Q_E = 71.4 + 50 =$ **121.4 kW** (pump work neglected)

> Cooling water needed for $\Delta T = 6$ K: $m = 121.4/(4.187\times6) =$ **4.83 kg/s**

Problem 2 (steam jet)

10 kg/s of water is chilled from 12 °C to 6 °C in a flash chamber ($h_{fg} = 2487$ kJ/kg). Entrainment ratio 0.4.

> Capacity = 10 × 4.187 × 6 = 251.2 kW = **71.4 TR** ; flash vapour = 251.2/2487 = 0.101 kg/s

> Motive steam = 0.101/0.4 = 0.2525 kg/s = **909 kg/h** → 12.7 kg/h per TR

Problem 3 (vortex tube)

Air at 400 kPa, 300 K enters; $_c = 0.3$, cold air leaves at 255 K. Find hot air temperature:

> $T_h = (300 - 0.3\times255)/0.7 = (300 - 76.5)/0.7 =$ **319.3 K**

Comparison of special refrigeration systems

System	Energy input	COP	Special feature
Three-fluid absorption	Heat (flame / heater)	0.2 - 0.3	No moving parts, H_2 balances pressure
Steam jet	High-pressure steam	0.2 - 0.4	Water refrigerant, ejector
Thermo-electric	DC electricity	0.3 - 0.8	Solid state, Peltier effect
Vortex tube	Compressed air	0.1 - 0.2	No moving parts, spot cooling

9. Quick Revision and Exam Questions

- $COP_{max} = (T_G - T_0)/T_G \cdot T_E/(T_0 - T_E)$; VAR uses heat; absorber + pump + generator replace the compressor.
- NH_3-water needs rectifier; LiBr-water does not (non-volatile) but needs vacuum, T > 0 °C.
- Three-fluid: NH_3 + water + H_2, no pump, same total pressure.
- Steam jet: nozzle – mixing – diffuser; water refrigerant; ≈ 10 kg steam/h per TR.
- Peltier: $Q_C = IT_C - ½I^2R - K\Delta T$. Vortex: $_cT_c + (1-_c)T_h = T_i$.

Short answers (2 marks)

- **What is the function of a rectifier?** Removes water vapour from NH_3 vapour so that only pure NH_3 enters the condenser.
- **Why no rectifier in LiBr system?** LiBr is non-volatile; only water vapour leaves the generator.
- **What is entrainment ratio?** Mass of vapour sucked per kg of motive steam.
- **Why is hydrogen used in Platen-Munters?** To equalise total pressure and lower the partial pressure of NH_3 so it can evaporate without a pump or valve.
- **What is Peltier effect?** Heating or cooling at the junction of two dissimilar conductors when current passes through.
- **Cold mass fraction?** Ratio of cold air mass flow to the total air flow in a vortex tube.
- **Why does LiBr plant need vacuum?** Water has to evaporate at about 4 °C, which needs 6 mm Hg absolute.

Likely questions

- Describe the NH_3-water absorption system with a neat sketch; role of rectifier, analyser. (long)
- Explain LiBr-water two-shell / four-shell systems; compare with NH_3-water. (long)
- Derive the expression for maximum COP of an absorption system. (long)
- Explain the three-fluid absorption refrigeration system. (long)
- Steam jet refrigeration: working, components, estimation of motive steam. (long)
- Thermo-electric refrigerator and vortex tube – working, advantages. (short / long)

Syllabus: Psychrometric properties and processes, sensible and latent heat loads, need for ventilation, infiltrated air, heat load concepts. Air cooler (evaporative cooling), window, split, summer, winter, year-round and central air conditioning systems.

I. Introduction to Air Conditioning

- **Air conditioning** = simultaneous control of temperature, humidity, air motion (velocity) and cleanliness (purity) of air in an enclosed space, to meet the requirement of occupants or of a process.
- **Comfort AC:** homes, offices, theatres, vehicles (≈ 22-26 °C, 40-60 % RH). **Industrial AC:** process control – textile mills, pharma, printing, computer rooms, operation theatres.
- Four basic functions: **cooling / heating, humidifying / dehumidifying, air cleaning (filtering), air distribution (and ventilation).**

Psychrometry – properties of moist air

Moist air = dry air + water vapour (Dalton's law: $p = p_a + p_v$). Properties used:

- **Dry bulb temperature (DBT)** – temperature by ordinary thermometer.
- **Wet bulb temperature (WBT)** – temperature of a wetted bulb in moving air; measures evaporative cooling potential. **WBT ≤ DBT.**
- **Dew point temperature (DPT)** – temperature at which vapour begins to condense when air is cooled at constant pressure; $p_{sat}(DPT) = p_v$.
- **Relative humidity ()** = p_v/p_{sat} at the DBT.
- **Specific (absolute) humidity (W)** = mass of vapour / mass of dry air.

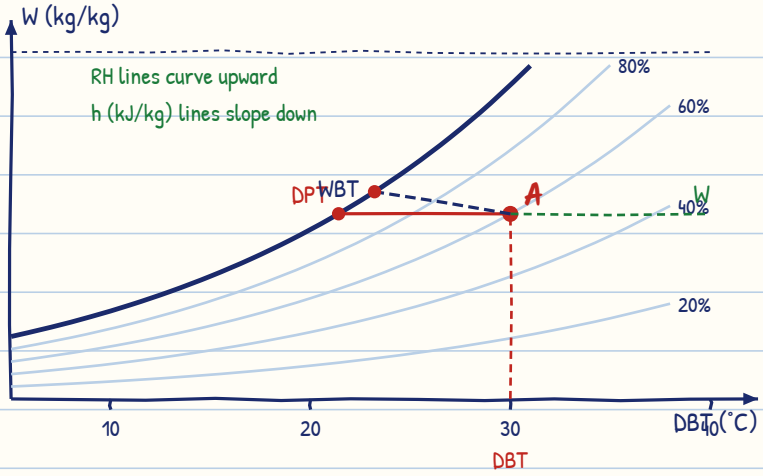
$$W = 0.622\ p_v / (p - p_v)$$

- **Degree of saturation** = W / W_s (at same DBT).
- **Specific enthalpy** of moist air per kg dry air:

$$h = 1.005\ t + W\ (2500 + 1.88\ t)\ kJ/kg\ d.a.$$

- **Specific volume** $v = R_a\ T\ (1 + 1.608\ W) / p \approx 0.83 - 0.95\ m^3/kg\ d.a.$

Note: For saturated air DBT = WBT = DPT. For unsaturated air DBT > WBT > DPT.



Point A: 30 °C DBT, 60 % RH
- horizontal line → DPT (21.4 °C)
- constant W line = dew point line
- sloping line up-left → WBT (≈ 23.2)
- enthalpy ≈ 71 kJ/kg
- v ≈ 0.88 m³/kg

Example 1 – properties by calculation

Air at 30 °C, 60 % RH, $p = 101.325$ kPa. $p_{sat}(30\ °C) = 4.246$ kPa.

$$p_v = 0.6 \times 4.246 = 2.548 \text{ kPa} \;;\; \text{DPT: } p_{sat} = 2.548 \rightarrow \mathbf{21.4\ °C}$$

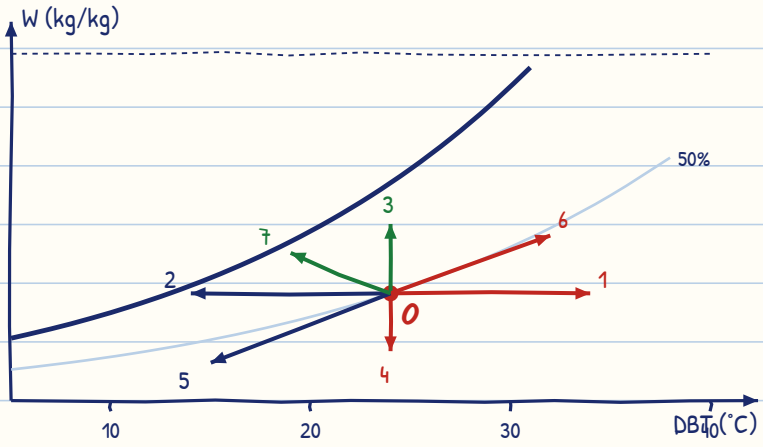
$$W = 0.622 \times 2.548 \,/\, (101.325 - 2.548) = \mathbf{0.01604\ kg/kg}$$

$$h = 1.005 \times 30 + 0.01604\,(2500 + 1.88 \times 30) = 30.15 + 41.0 = \mathbf{71.2\ kJ/kg}$$

$$v = 0.287 \times 303\,(1 + 1.608 \times 0.01604)/101.325 = \mathbf{0.88\ m^3/kg}$$

2. Psychrometric Chart and Processes

- The **psychrometric chart** plots W (vertical, right) against DBT (horizontal). Lines on it: constant DBT (vertical), constant W (horizontal), constant RH (curves), constant WBT (sloping down to right), constant enthalpy (nearly same as WBT), constant specific volume (steeper slope).
- Any two independent properties fix the state.



1 sensible heating

2 sensible cooling

3 humidification

4 dehumidification

5 cooling + dehumidification

6 heating + humidification

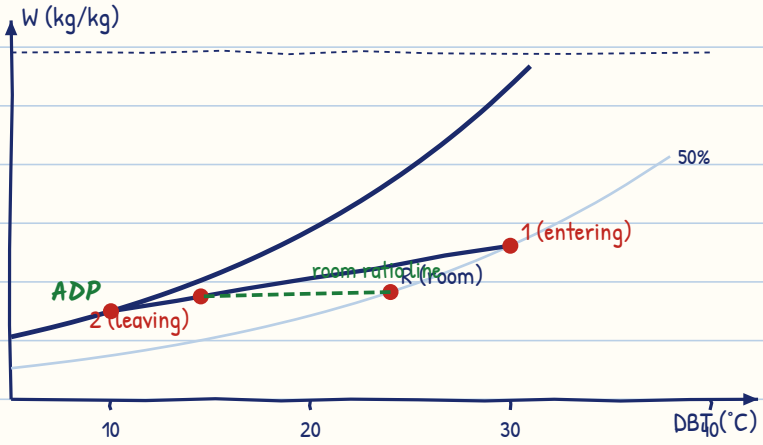
7 evaporative cooling (WBT)

+ mixing: point on line joining

Process equations (m_a = dry air flow, kg/s)

- **Sensible heating / cooling** (W = const): $Q = m_a\, c_{pm}\, \Delta t = m_a\,(1.005 + 1.88\,W)\,\Delta t \approx 1.022\, m_a\, \Delta t$ kW. With volume flow V (m^3/s): **$Q_s = 1.23\, V\, \Delta t$** kW.
- **Humidification** (heating and adding steam): latent heat $Q_L = m_a\,(W_2 - W_1)\,h_{fg}$; **$Q_L = 3012\, V\, \Delta W$** kW ($h_{fg} \approx 2500$, $\rho = 1.2$).
- **Cooling and dehumidification:** air cooled below its dew point; $Q = m_a\,(h_1 - h_2) - m_w\, h_f$; condensate $m_w = m_a\,(W_1 - W_2)$.
- **Heating and humidification (winter AC):** $Q_{heat} = m_a\,(h_2 - h_1)$.
- **Adiabatic mixing of two streams:** $h_3 = (m_1h_1 + m_2h_2)/(m_1 + m_2)$, $W_3 = (m_1W_1 + m_2W_2)/(m_1 + m_2)$ – the mixed state lies on the straight line joining 1 and 2.
- **Adiabatic saturation / evaporative cooling:** h ≈ constant along WBT line; DBT falls, W rises.
- **Chemical dehumidification:** air passes over silica gel / calcium chloride – W falls, DBT rises (h ≈ const).

Cooling coil – ADP and bypass factor



ADP = coil surface temp.

Bypass factor

$BF = (t_2 - t_{ADP}) / (t_1 - t_{ADP})$

Contact factor = 1 – BF

SHF = SH / (SH + LH)

- **Apparatus Dew Point (ADP):** the average coil surface temperature. Air leaving a coil is a mix of air that touched the coil (at ADP) and air that bypassed.

$$BF = (t_2 - t_{ADP}) / (t_1 - t_{ADP}) \text{ ; Contact factor } CF = 1 - BF$$

- BF is smaller for deeper coils, more fins, lower air velocity. Typical BF: 0.1-0.3.
- **Sensible Heat Factor:** $SHF = Q_s / (Q_s + Q_L)$. **RSHF** (room), **GSHF** (grand = room + ventilation load), **ESHF** (effective, at coil).
- The **room ratio line** (slope fixed by RSHF) is drawn from the room state; its intersection with the saturation line gives the required ADP.

How to plot a state on the chart (steps)

- Mark the DBT on the horizontal axis and go vertically up; read the intersection with the RH curve (or WBT line) – that is the state point.
- From the point go horizontally to the right to read W; go along the sloping line to the left to read WBT / enthalpy scale; horizontally left to the saturation curve for DPT.
- For a process draw a straight line between the two states; use **protractor (SHF scale)** on the chart for the room ratio line.
- Condition lines: comfort zone ≈ 22-26 °C DBT and 40-60 % RH (the "bounded zone").

3. Sensible and Latent Heat Loads

Sensible heat loads

- **Solar radiation** through windows (largest): $Q = A \times SHGF \times SC$.
- **Transmission** through walls, roof, floor, glass: $Q = U\ A\ (CLTD)$ or $U\ A\ (t_o - t_i)$.

- **Internal:** lights (W × 1.25 for fluorescent ballast), people (≈ 75 W sensible per person seated), motors and equipment, cooking appliances.
- **Infiltration and ventilation** air (sensible portion): $Q_s = 1.23\ V\ (t_o - t_i)$.

Latent heat loads

- **People** (≈ 55 W latent per person), **infiltration / ventilation** air moisture: $Q_L = 3012\ V\ (W_o - W_i)$, cooking, steam from processes, evaporation from wet surfaces, door openings.

Source	Sensible	Latent
Occupants (seated)	75 W/person	55 W/person
Fluorescent lights	1.25 × W	–
Equipment (computers)	rated W × use factor	–
Walls, roof, glass	U A ΔT, solar	–
Infiltration / ventilation	1.23 V Δt	3012 V ΔW
Cooking / steam	some	large

Solar and transmission load formulae

Solar gain through glass: $Q = A \times SHGF_{max} \times SC \times CLF$ (A = glass area, SC = shading coefficient)

Wall / roof: $Q = U\ A\ (CLTD)$; $U = 1 / (1/h_o + \Sigma\ L/k + 1/h_i)$ W/m^2K

- Typical U: brick wall 2.0, insulated roof 0.4–0.6, single glass 5.8, double glass 2.8. CLTD (cooling load temperature difference) accounts for time lag due to thermal storage.
- **Diversity factor:** not all loads peak together; people, lights → multiply by use factor 0.7–0.9.

Total load

Room sensible heat (RSH) + room latent heat (RLH) = Room total heat (RTH); RSHF = RSH / RTH

Grand total heat = RTH + outdoor air load (OASH + OALH) + supply duct gain, fan heat, leakage.

- Add 5–10 % safety factor. Cooling TR = Grand total heat (kW)/3.5167.

4. Need for Ventilation and Infiltrated Air

Ventilation

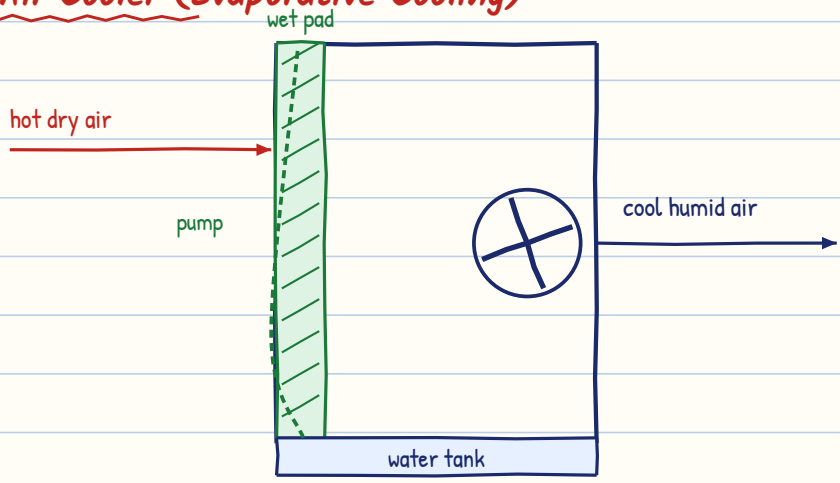
- **Needed:** to supply oxygen, dilute CO_2 (limit ≈ 1000 ppm) and body odours, smoke, VOCs; to maintain positive pressure in clean rooms.
- Fresh air is mixed with the return air; typical requirement ≈ 7-10 L/s (≈ 0.4-0.6 m^3/min) per person for offices; ASHRAE 62.1 ≈ 2.5 L/s per person + 0.3 L/s per m^2 floor.
- **Fresh air = 10-30 %** of supply air. Too much fresh air adds load; too little affects health ("sick building").
- Methods: natural (windows, stack), mechanical (supply fan, exhaust fan), heat recovery wheel (ERV) to save energy.

Infiltration

- Uncontrolled leakage of outside air through cracks around doors and windows, door openings, and the **stack effect** (hot air rises) and **wind pressure**.
- **Air change method:** V = (ACH × room volume)/3600 m^3/s. Typical 0.5-1 ACH for houses, 1-2 for shops with door use.
- **Crack method:** V = (crack length) × (leakage per m) at the design pressure difference.
- Reduce by weather stripping, double doors, air curtain, and keeping the room under slight positive pressure.

Infiltration load: $Q_s = 1.23\ V\ (t_o - t_i)$, $Q_L = 3012\ V\ (W_o - W_i)$

5. Air Cooler (Evaporative Cooling)



Desert cooler (direct)

Process: sensible → latent

$t_2 = t_1 - \eta\ (t_1 - t_{wb})$

η = 70-90 %

DBT ↓ , W ↑ , h ≈ const

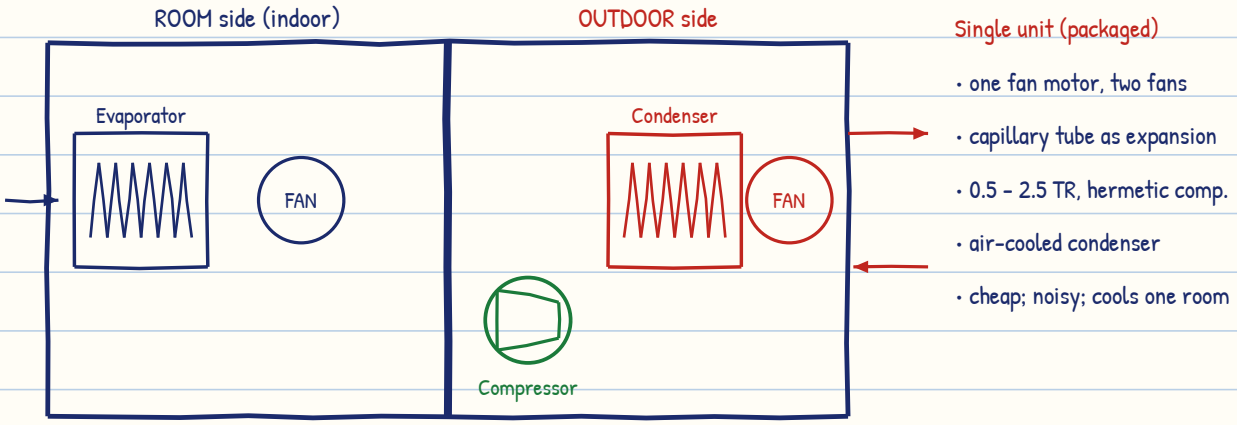
- Water is circulated by a small pump over a porous pad (khus / honeycomb cellulose). Warm dry air passing over the wet pad evaporates water, so the **air loses sensible heat** to supply latent heat – DBT falls, humidity rises (h ≈ const).

> Saturation effectiveness $\eta = (t_1 - t_2) / (t_1 - t_{wb})$

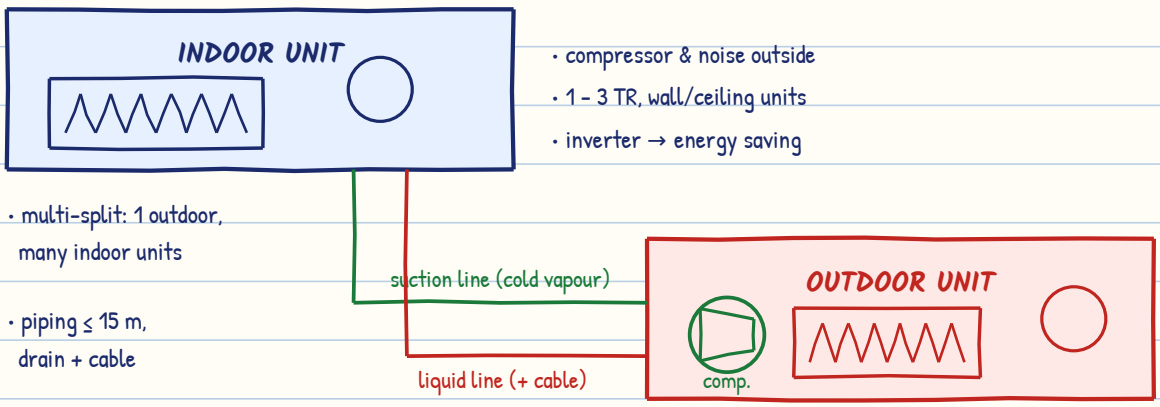
- Example: outside 40 °C, WBT 24 °C, $\eta = 0.8 \rightarrow t_2 = 40 - 0.8 \times 16 =$ **27.2 °C**.
- **Suitable for hot and dry climates** (Rajasthan, Deccan); poor in humid coastal regions. Low cost (power ≈ 1/10 of AC), needs ventilation (open window) to exhaust humid air. **Indirect** evaporative cooler uses a heat exchanger – no humidity addition. **Two-stage** (indirect + direct) can cool below WBT.

6. Types of Air Conditioning Systems

Window air conditioner

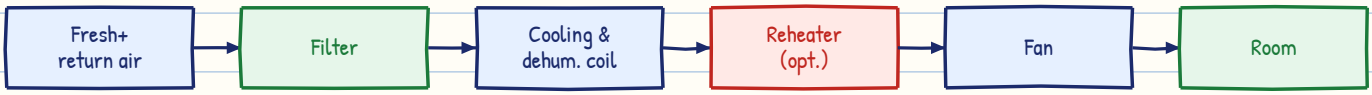


Split air conditioner

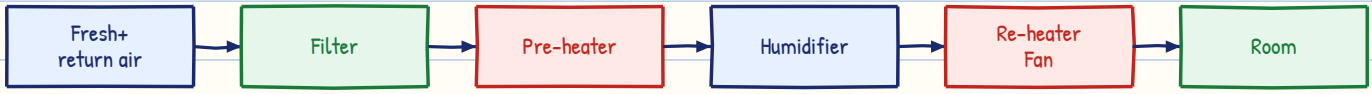


Summer, Winter and Year-round systems

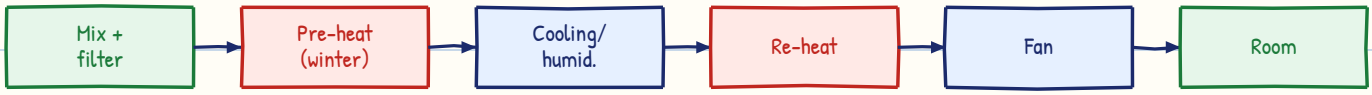
Summer AC



Winter AC

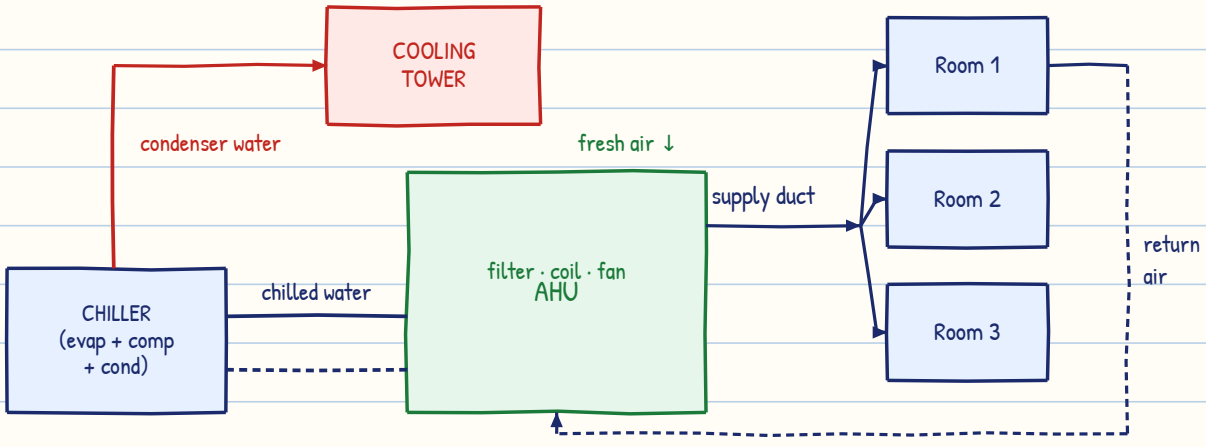


Year-round



- **Summer AC:** cooling and dehumidification with the cooling coil, then reheating if needed. Typical room 24-26 °C and 50 % RH. Load: RSHF line and ADP.
- **Winter AC:** heating and humidification (preheat coil → humidifier / steam sprayer → reheat coil). Room 21-22 °C, 40 % RH. Pre-heating avoids freezing of water spray.
- **Year-round AC:** both cooling/dehumidifying and heating/humidifying coils in one plant, with controls for the season, fresh-air mixing and reheat. Needed in places with both hot and cold seasons.

Central air conditioning



- Large buildings, hotels, hospitals, malls: **chiller plant** (VCR or VAR) makes chilled water (7 → 12 °C) circulated to **Air Handling Units (AHU)** or fan coil units; AHU conditions air and ducts distribute it. Heat rejection through a **cooling tower**.

- **Types:** all-air (constant air volume, VAV), air-water (fan coil + primary air), all-water, **VRF/VRV** (variable refrigerant flow).
- **Pros:** centralised maintenance, better humidity control, one plant for many zones, quiet in rooms. **Cons:** high initial cost, large duct space, skilled operators needed.

Type	Capacity	Pros	Cons
Window	0.5 – 2.5 TR	Cheap, simple	Noisy, one room
Split	1 – 5 TR	Quiet, flexible	Pipe length limit
Packaged	3 – 25 TR	Ducted, easy install	Space outside
Central	> 25 TR	Large buildings, control	Costly, ducts

6A. Air Distribution, Ducts and AHU

- **Supply outlets:** grille, register (with damper), ceiling diffuser (square / round / linear slot). Aim: uniform temperature, velocity < 0.25 m/s in occupied zone (draught-free). **Throw** = distance air travels from outlet to terminal velocity (0.25 m/s); **drop**, **spread**.
- **Return air** through grilles at low level/ceiling. Supply-return balanced so that room pressure is slightly positive.
- **Duct design:** equal friction method (≈ 0.8-1.0 Pa/m), velocity reduction, static regain; main duct velocity 6-9 m/s, branches 3-5 m/s; rectangular / round GI sheet with insulation and vibration-free flexible connections.

AHU components

- Mixing box with dampers (fresh + return) → pre-filter → cooling coil (chilled water / DX) → heating coil → humidifier → supply fan (centrifugal) → filters (fine) → supply duct. Drain tray for condensate.
- **Fan coil unit (FCU):** small coil + fan in the room, chilled / hot water supply; **VAV:** supply volume varies with load (saves fan energy); **VRF:** refrigerant flow varied by inverter compressor to many indoor units.

Direct expansion (DX)	Chilled water
Refrigerant circulated to coil	Water circulated, primary VCR/VAR at plant
Up to ≈ 25 TR, packaged	Large buildings, > 25 TR
Short pipe, cheap	Long distance, better control

Controls

- Thermostat (DBT), humidistat (RH), pressure cut-outs, damper actuators; BMS for large buildings. Set-points: 24 ± 1 °C and 50 ± 5 %.

Step-wise design procedure for summer air conditioning

- **Step 1:** fix outdoor design (e.g. 35 °C DBT, 24 °C WBT) and indoor design (24 °C, 50 %) conditions.
- **Step 2:** calculate room sensible (RSH) and latent (RLH) loads → RSHF.
- **Step 3:** draw the room ratio line through the room state R; choose supply temperature difference (8-12 K) → supply state S; or find ADP where the line cuts the saturation curve.
- **Step 4:** supply air $V = RSH / (1.23 (t_R - t_S))$. Fresh air fraction chosen from the ventilation rule; mix with return air to get state M.
- **Step 5:** coil enters at M, leaves at S (along the line to ADP): find BF; coil load = $m_a (h_M - h_S)$ → TR.
- **Step 6:** select fan, ducts, filters, chiller, cooling tower; add reheater if S is too cold.

Space	Fresh air (L/s per person)	Design condition
Office	7 - 10	24 °C, 50 %
Theatre	5 - 8	24 °C, 55 %
Restaurant	10 - 15	24 °C, 55 %
Hospital ward	12 - 15	23 °C, 50 %
Operation theatre	100 % fresh, 20 ACH	21 °C, 50 %

Energy-saving measures

- Use inverter compressors, VAV and VRF; energy recovery ventilators; economiser (free cooling using cool outside air); insulation, double glazing, shading; set point 24-26 °C; clean filters and coils.

7. Numerical Problems

Problem 1 (sensible heating)

2 m^3/s of air is heated from 10 °C to 30 °C. $Q = 1.23 \times 2 \times 20 =$ **49.2 kW**.

Problem 2 (mixing)

2 kg/s air at 25 °C, $W = 0.010$ is mixed with 1 kg/s at 35 °C, $W = 0.020$.

$t_3 = (2\times25 + 1\times35)/3 = \mathbf{28.3\ °C}$; $W_3 = (2\times0.010 + 1\times0.020)/3 = \mathbf{0.0133\ kg/kg}$

Problem 3 (supply air quantity)

Room 24 °C, 50 % RH, RSH = 20 kW and RLH = 5 kW → RSHF = 20/25 = **0.8**. Supply air at 14 °C:

$$V = RSH / (1.23\ \Delta t) = 20 / (1.23 \times 10) = \mathbf{1.63\ m^3/s} \approx 5850\ m^3/h$$

Problem 4 (bypass factor)

Coil entering air 28 °C, leaving 14 °C, ADP 10 °C: BF = (14 − 10)/(28 − 10) = **0.222**, CF = 0.778.

Problem 5 (heat load of a 100 m² office, 20 persons)

People: sensible 20×75 = 1.5 kW ; latent 20×55 = 1.1 kW

Lights 1.0 kW × 1.25 = 1.25 kW ; Equipment 2 kW ; Envelope 5 kW

Fresh air: 20 × 7.5 L/s = 0.15 m³/s ; 35 °C → 24 °C, ΔW = 0.0075

OA sensible = 1.23×0.15×11 = 2.03 kW ; OA latent = 3012×0.15×0.0075 = 3.39 kW

Total sensible = 1.5+1.25+2+5+2.03 = 11.78 kW ; Total latent = 1.1+3.39 = 4.49 kW

Grand total = 16.27 kW = **4.6 TR** ; SHF = 11.78/16.27 = 0.72

Problem 6 (evaporative cooler)

t_1 = 38 °C, t_{wb} = 22 °C, η = 85 % → $t_2 = 38 - 0.85 \times 16 = \mathbf{24.4\ °C}$.

Problem 7 (cooling coil load)

1.5 kg/s of air enters a coil at h_1 = 60 kJ/kg, W_1 = 0.0140 and leaves at h_2 = 40 kJ/kg, W_2 = 0.0090.

Condensate = 1.5 (0.0140 − 0.0090) = 0.0075 kg/s = **27 kg/h**

$Q = 1.5\ (60 - 40) - 0.0075 \times 58.8 = 30 - 0.44 =$ **29.6 kW = 8.4 TR**

Problem 8 (ventilation load of a theatre)

200 persons at 7.5 L/s = 1.5 m^3/s outdoor air, 35 °C to 24 °C, $\Delta W = 0.0075$:

OA sensible = 1.23×1.5×11 = 20.3 kW ; OA latent = 3012×1.5×0.0075 = 33.9 kW ; total = **54.2 kW = 15.4 TR**

Note: Ventilation load is often 25–40 % of the total – energy recovery wheel is used to reduce it.

Problem 9 (infiltration)

Room 5 × 4 × 3 = 60 m^3 with 1.5 ACH → V = 60×1.5/3600 = 0.025 m^3/s.

$Q_s = 1.23 \times 0.025 \times 11 = 0.34$ kW ; $Q_L = 3012 \times 0.025 \times 0.0075 = 0.56$ kW ; total = **0.9 kW**

Short answers (2 marks)

- **What is dew point?** The temperature at which water vapour starts condensing when air is cooled at constant pressure and constant W.
- **Why is SHF important?** It fixes the slope of the room ratio line and the ADP.
- **What is infiltration?** Uncontrolled leakage of outside air into the conditioned space.
- **Evaporative cooling – process line?** Along constant WBT / constant enthalpy.
- **Which AC for large hotels?** Central AC with chiller, AHU and cooling tower.

8. Quick Revision and Exam Questions

- $W = 0.622\ p_v/(p - p_v)$; $h = 1.005t + W(2500 + 1.88t)$; $Q_s = 1.23\ V\ \Delta t$; $Q_L = 3012\ V\ \Delta W$.
- SHF = SH/(SH+LH); BF = $(t_2 - t_{ADP})/(t_1 - t_{ADP})$; evaporative $\eta = (t_1 - t_2)/(t_1 - t_{wb})$.
- Window = packaged; split = indoor + outdoor; central = chiller + AHU + cooling tower.

Likely questions

- Define DBT, WBT, DPT, RH, specific humidity; show on psychrometric chart. (short)
- Explain psychrometric processes with chart and equations. (long)
- Define SHF, RSHF, GSHF, ADP and bypass factor. (short / long)
- Components of cooling load; need for ventilation and infiltration. (long)

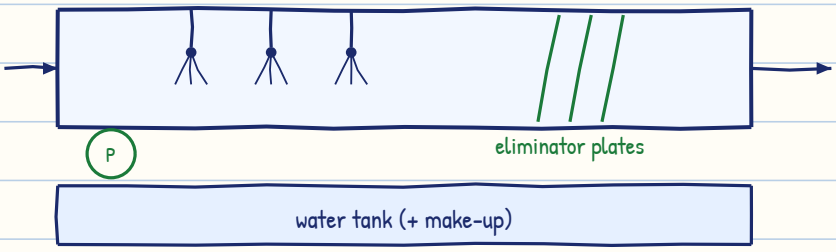
- Explain with schematic: summer, winter, year-round AC. (long)
- Window, split and central AC – compare. (long)
- Evaporative cooling – principle, effectiveness, application. (short)

Syllabus: Air conditioning equipment – humidifiers, dehumidifiers, air filters, fans and blowers. Human comfort – requirements of temperature and humidity, effective temperature, comfort chart. Heat pump – heat sources, different heat pump circuits. Energy efficient AC systems – star rating.

I. Humidifiers

- A **humidifier** adds moisture to air (increases W) – needed in winter air-conditioning, textile mills, printing, paper stores, hospitals, and museums.

(a) Air-washer / spray humidifier

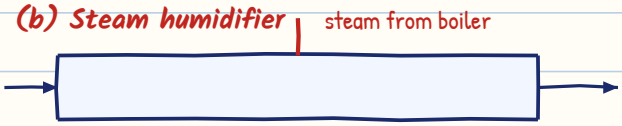


Spray: water droplets evaporate in air → W ↑, DBT ↓ (adiabatic)

Steam: W ↑, DBT ≈ const (isothermal), hygienic

$m_s = m_a (W_2 - W_1)$

(b) Steam humidifier



Types

- **Spray / air washer:** water sprayed by nozzles into the air stream; evaporative (adiabatic) – air cools while W rises. Recirculating pump, tank, eliminators to catch droplets. Also cleans the air.
- **Steam humidifier:** steam from a boiler or electrode / resistance boiler injected into the duct. Almost **isothermal**, hygienic, accurate control; costly energy.
- **Evaporative pad / pan humidifier:** wetted pad, air blown across; used in small units.
- **Ultrasonic humidifier:** piezo-crystal vibrates at 1.7 MHz, breaking water into micro-droplets (mist). Low power; needs demineralised water.
- **Capillary / wetted-media, centrifugal atomising** – domestic types.

Steam needed $m_s = m_a (W_2 - W_1)$ kg/s ; heat added ≈ m_s × 2676 kJ/kg (for steam at 100 °C)

Example

2 kg/s of air is humidified from W = 0.006 to 0.009: m_s = 2 × 0.003 = **0.006 kg/s = 21.6 kg/h** and heat = 0.006 × 2676 = **16 kW**.

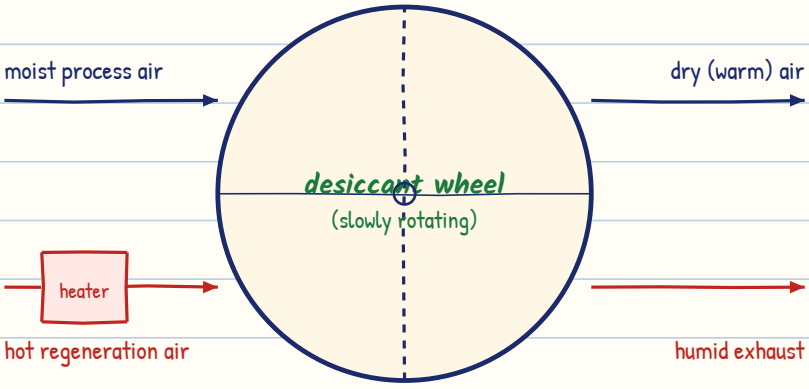
Note: Humidity control by humidistat; keep RH > 30 % in winter (static electricity, dry skin) and < 60 % (mould).

2. Dehumidifiers

- Removing moisture (reducing W) is needed in humid climates, cold storage, pharma, electronics, ships, basements and where closely controlled RH is required.

Methods

- **Cooling and dehumidification:** pass air over cooling coil below its DPT – water condenses (used in all ACs and **refrigerant dehumidifiers**). Air is then **reheated** if needed. Low energy at moderate humidity, poor below ≈ 40 % RH.
- **Chemical (sorption) dehumidification:** hygroscopic material absorbs / adsorbs the vapour.
  - **Absorbent (liquid):** lithium chloride, triethylene glycol sprays – the vapour is absorbed; liquid regenerated by heating.
  - **Adsorbent (solid):** silica gel, activated alumina, molecular sieve, zeolite – vapour held on surface; regenerated by hot air (100-130 °C).
- **Compression:** compressing the air increases p_v → water condenses (compressed air driers).



Adsorption on silica gel, molecular sieve or LiCl.
Process side: W ↓, DBT ↑
Regeneration: 60 – 130 °C
hot air drives off moisture

Method	Process line	Remarks
Cooling coil	W ↓ and DBT ↓	Common, needs reheat
Chemical (adsorption)	W ↓, DBT ↑ (h ≈ const)	Very dry air, 10 % RH possible
Desiccant wheel + cooling	Hybrid	Saves energy, uses waste heat

3. Air Filters

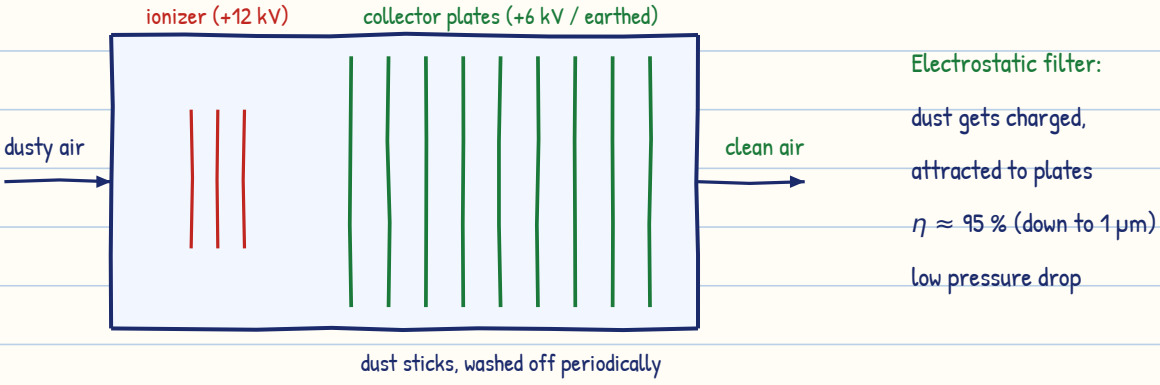
- Purpose: remove **dust, pollen, smoke, bacteria, fungus, odour** from supply / return air – to protect health, occupants, equipment and process.

Terms

- **Efficiency** – percentage of particles of a given size removed. **Arrestance** – weight of arrested dust. **Dust-holding capacity, pressure drop** (initial 50-100 Pa → final 250 Pa; change filter), **MERV rating (1-16)**.

Types of air filters

- **Dry (fibre / panel) filters:** glass fibre, cotton, synthetic mat, pleated paper; throw-away or washable; cheap, pre-filter (efficiency 20-80 % at > 5 m). Face velocity 1-2.5 m/s.
- **Viscous-impingement (oil-coated) filters:** metal mesh or foam coated with oil; dust sticks to the oil film; washed and re-oiled; used in industrial / kitchen.
- **Wet filters (air washers):** water spray / scrubber – removes dust and soluble gases; also humidifies.
- **Electrostatic precipitator (ESP):** dust particles charged by corona, collected on oppositely charged plates; very high efficiency, low pressure drop, high initial cost and ozone risk.



- **HEPA (High Efficiency Particulate Air) filter:** glass micro-fibre pleated paper, **99.97 % at 0.3 m**. Cleanrooms, hospital operating theatres, pharma, labs. ULPA is 99.999 % at 0.12 m.

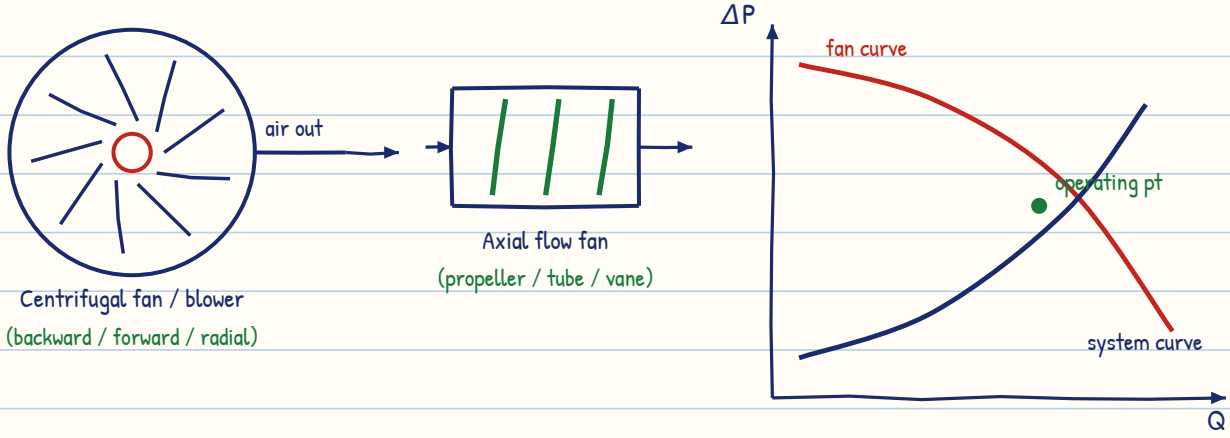
- **Activated carbon filters:** adsorb odours, gases, VOCs (kitchen, smoke).

Class	Typical efficiency	Use
Pre-filter (G3-G4)	60-90 % (> 10 m)	All AHUs, first stage
Fine filter (F7-F9)	80-95 % (1 m)	Offices, hospitals
HEPA (H13/H14)	99.95-99.995 % (0.3 m)	Cleanrooms, O.T.
Electrostatic	90-98 %	Industrial, hospitals

Note: Filter staging: pre-filter → fine filter → HEPA (final). Never skip the pre-filter – it protects the costly filters.

4. Fans and Blowers

- Fans supply air through ducts with small pressure rise (up to ≈ 1.1 pressure ratio, ~ 1-3 kPa); **blowers** give higher pressure (up to ≈ 10-15 kPa); compressors beyond.



Classification

- **Centrifugal (radial-flow):** air enters axially and leaves radially. Blades: **forward-curved** (small, quiet, low speed, used in FCU, AHU, power ↑ with flow – overloading), **backward-curved / airfoil** (highest efficiency, non-overloading, large AHU), **radial** (rugged, dusty air).
- **Axial flow:** air flows parallel to the shaft. **Propeller** (low pressure, large flow – ventilation), **tube-axial** (with duct casing), **vane-axial** (with guide vanes, high efficiency).
- **Cross-flow (tangential):** long cylindrical impeller – indoor split AC units, quiet.

Fan type	Flow	Pressure	Efficiency	Use
Propeller	Very high	Very low	40-50 %	Exhaust, cooling tower
Tube / vane-axial	High	Medium	70-85 %	Duct ventilation
Forward-curved	Medium	Medium	60-70 %	FCU, packaged AC
Backward-curved	High	High	80-90 %	Large AHU

Fan performance

Air power = $Q\,\Delta P$; Shaft power $P = Q\,\Delta P / \eta_{fan}$ (Q in m^3/s, ΔP in Pa → W)

Fan laws (same fan, speed N): $Q \propto N$, $\Delta P \propto N^2$, Power $\propto N^3$

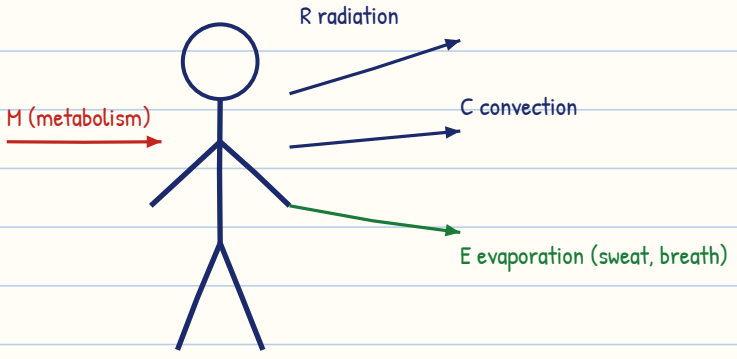
- Operating point is the intersection of **fan curve** and **system resistance curve** ($\Delta P \propto Q^2$). Control by dampers, inlet guide vanes or VFD (variable frequency drive – most energy efficient).

Example

- $Q = 2\ m^3/s$, $\Delta P = 500$ Pa, $\eta = 0.65$ → $P = 2\times500/0.65 =$ **1538 W ≈ 1.54 kW**.
- A fan gives $5\ m^3/s$, 400 Pa at 1000 rpm, 3 kW. At 1200 rpm: $Q = 6\ m^3/s$, $\Delta P = 400\times1.44 = 576$ Pa, power = $3\times1.728 =$ **5.18 kW**.

5. Human Comfort

- **Thermal comfort** is the condition of mind that expresses satisfaction with the thermal environment (ASHRAE 55). The body produces heat by metabolism (≈ 100 W for a seated person, 1 met = 58 W/m^2) and must lose it at the same rate to keep deep body temperature ≈ 37 °C.



Heat balance of body:

M - W = E + R + C + S

S = 0 → thermal comfort

S > 0 : feels hot

S < 0 : feels cold

- **Heat losses:** convection (air movement), radiation (to cooler surfaces), evaporation (sweat and respiration), conduction (negligible). Higher DBT and humidity → less loss → feels hot.

Factors affecting comfort

Factor	Effect
Air temperature (DBT)	Main; 22-26 °C comfortable
Humidity (RH)	High RH reduces sweat evaporation; optimum 40-60 %
Air velocity	0.1-0.25 m/s gentle; > 0.5 m/s draught; helps in summer
Mean radiant temperature	Radiation from walls, sun
Clothing (clo) and activity (met)	1 clo = 0.155 m^2K/W
Air quality, noise, light	Fresh air and cleanliness

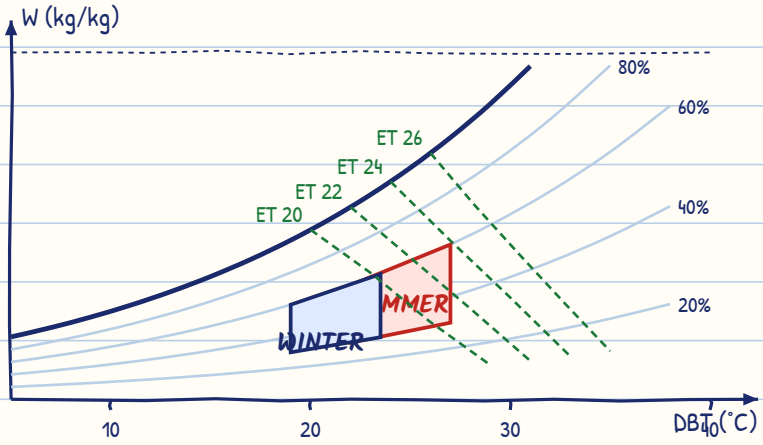
Requirements of temperature and humidity

- **Summer:** 24-26 °C DBT, 50-60 % RH (a little warmer to avoid too large indoor / outdoor difference – about 7-8 K). **Winter:** 20-22 °C, 40-50 % RH.
- Low RH (< 30 %) → dry mucous membranes, static; high RH (> 70 %) → clammy feeling, mould. Air velocity 0.15-0.25 m/s, vertical temperature difference < 3 K, fresh air.

Effective Temperature (ET)

- **Definition:** an index that combines the effects of DBT, humidity and air motion – ET is the temperature of **still, saturated air (RH = 100 %, V ≈ 0)** which produces the same sensation of warmth or cold as the given air state.
- Example: air at 26 °C and 50 % RH in still air has ET ≈ 22.5 °C → feels like saturated air at 22.5 °C.
- **Drawbacks:** over-estimates the effect of humidity at low temperatures; does not include radiation (corrected ET with globe temperature) and clothing/activity.

Comfort chart



Comfort chart (ASHVE)

ET lines slope down-right

(sat. air, still = ET)

Summer zone: ET ≈ 22–26 °C

Winter zone: ET ≈ 19–23 °C

RH 30 – 70 %

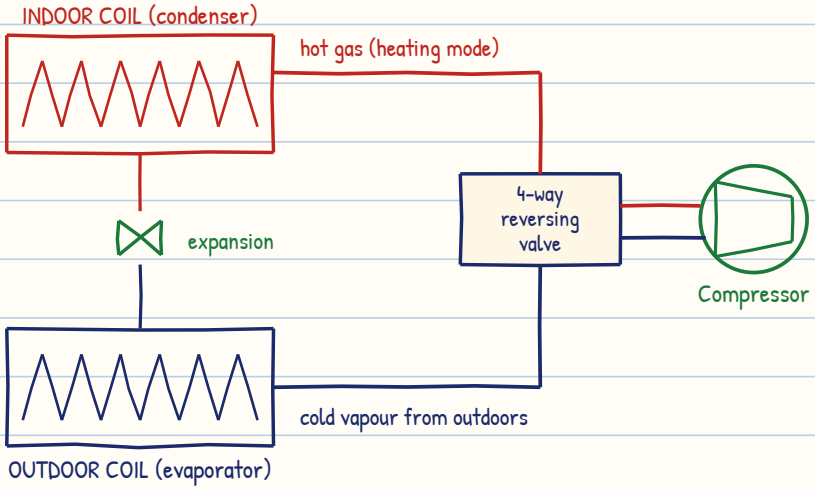
- The **ASHVE comfort chart** (Houghton and Yaglou) is the psychrometric chart on which lines of constant ET are plotted; the **comfort zones** for summer and winter, where ≥ 50 % of the persons are comfortable, are drawn.
- Using the chart: given DBT and RH → read ET; or choose the ET (say 23 °C) and find different combinations of DBT/RH giving the same comfort – helpful for choosing the AC design point.
- **Other indices:** Predicted Mean Vote (**PMV**, -3 to +3) and Predicted Percentage Dissatisfied (**PPD** ≥ 5 %) by Fanger; operative temperature; heat stress index (HSI); wet-bulb globe temperature (WBGT).

6. Heat Pump

- A heat pump is a refrigerating machine used for **heating**: it extracts heat Q_L from a low-temperature source (outside air, ground, water) and delivers $Q_H = Q_L + W$ to the space.

$$COP_{HP} = Q_H / W = COP_R + 1 \geq 1 \; ; \; \text{Carnot } COP_{HP} = T_H / (T_H - T_L)$$

- Typical COP 3–5 (more than 1 kW electric heating) – hence energy saving and lower CO_2.



Heating: indoor coil = condenser

Cooling: valve reverses, indoor coil = evaporator

$COP_{HP} = Q_H/W = COP_R + 1$

Defrost cycle for outdoor coil

- **Reversing valve** (4-way) interchanges the roles of the coils: in cooling mode indoor coil is the evaporator, outdoor coil the condenser. **Defrost** cycle is needed for outdoor coils at < 5 °C.

Heat sources (low-grade heat)

Source	Temperature	Merits	Demerits
Atmospheric air	−10 to 15 °C	Free, available anywhere	COP drops in cold; frost
Well / ground water	8–15 °C steady	High COP, steady	Needs wells, legal permission
Ground (soil) – horizontal / vertical loops	5–12 °C	Steady, best COP	High installation cost
Surface water (lake, river)	2–15 °C	Easy	Freezing risk
Solar (solar-assisted)	variable	Free energy	Needs storage
Waste heat (industry, exhaust air)	20–40 °C	Very high COP	Site specific

Heat pump circuits (classification by source and distribution)

Circuit	Source	Distribution	Example
Air-to-air	Outdoor air	Room air (ducts)	Reversible split AC, VRF
Air-to-water	Outdoor air	Hot water (radiators)	Domestic hot water, hydronic
Water-to-air	Water / ground loop	Room air	Ground source unit
Water-to-water	Water / ground	Hot water	Large buildings
Earth-to-air / earth-to-water	Soil	Air / water	Geothermal heat pump

- **Other cycles:** absorption heat pump (heat driven; gas fired), **heat pump water heater**, **heat recovery chiller** (simultaneous cooling and heating), **cascade** systems.

Example

Heating load 12 kW, COP = 4: W = 12/4 = **3 kW**, heat from source = 12 - 3 = **9 kW**. Carnot COP_{HP} between 0 °C and 22 °C = 295/22 = 13.4; actual COP 4 = 30 % of Carnot.

7. Energy Efficient AC Systems and Star Rating

- AC use is a huge part of peak electricity demand in India, so **BEE (Bureau of Energy Efficiency)** runs the **Standards & Labelling** programme with star labels (1 star – 5 stars: more stars = more efficient).
- Rating metric: **ISEER** (Indian Seasonal Energy Efficiency Ratio) = total seasonal cooling delivered (Wh) / total electrical energy consumed (Wh) over a typical Indian cooling season at weighted outdoor temperature bins (24-43 °C). It includes part-load operation, so **inverter ACs get a higher rating**.

> ISEER = Seasonal cooling output / seasonal electricity input (energy consumed ∝ 1/ISEER)

BEE star bands (split AC, valid 1 Jan 2026 – 31 Dec 2028)

Star	ISEER (split AC)	Window AC ISEER
1 star	3.50 - 3.79	2.90 - 3.09
2 star	3.80 - 4.29	3.10 - 3.29
3 star	4.30 - 4.99	3.30 - 3.49
4 star	5.00 - 5.59	3.50 - 3.69
5 star	≥ 5.60	≥ 3.70

Note: BEE revises these bands every few years (the 2026 update was tougher: an old 5-star is now about a 4-star). Always check the label year and ISEER value; verify the latest on the BEE website.

Features of energy efficient AC

- **Inverter (variable-speed) compressor** – matches capacity to the load; saves 30-50 % energy vs fixed-speed. **BLDC fans**, large copper heat-exchanger area, **R32 / R290** low-GWP refrigerant, electronic expansion valve, smart sensors, Wi-Fi, auto clean, dry mode.
- **Operational tips:** set the temperature at 24-26 °C (each 1 °C higher saves ≈ 6 % energy), regular filter cleaning, shading and insulation, correct tonnage, fans to circulate air, use timers.
- **System-level:** VRF, variable air volume, chilled-water plants with VFD pumps / fans, cooling-tower optimisation, thermal storage, heat recovery, building management system (BMS), evaporative pre-cooling of condenser air.

Example

Replacing a 3.6 ISEER AC by a 5.6 ISEER AC for the same cooling duty: energy saving = 1 - 3.6/5.6 = **35.7 %**. If the old unit consumed 1500 kWh per year, the new one uses about 964 kWh – saving 536 kWh.

7A. Numerical Problems and Short Answers

Problem 1 (humidifier with spray)

Air at 25 °C, W = 0.008 passes an air washer with 80 % efficiency and leaves almost saturated. Saturated W at the WBT ≈ 0.0125. Then W_2 = 0.008 + 0.8 (0.0125 - 0.008) = **0.0116 kg/kg**. For 3 kg/s of air, water evaporated = 3 × 0.0036 = **0.0108 kg/s = 38.9 kg/h**.

Problem 2 (dehumidification by cooling)

Air 1.2 kg/s at W_1 = 0.016 is cooled to W_2 = 0.009. Condensate = 1.2 × 0.007 = **0.0084 kg/s** ≈ **30 kg/h**. Latent load = 0.0084 × 2450 = **20.6 kW**.

Problem 3 (filter efficiency)

Upstream dust 20 mg/m^3, downstream 1 mg/m^3: efficiency = (20 - 1)/20 = **95 %**.

Problem 4 (body heat balance)

Seated person produces M = 104 W, does no external work (W = 0). Evaporation (E) = 25 W, convection (C) = 35 W, radiation (R) = 40 W → S = 104 - 100 = **+4 W** (slightly warm, body stores heat).

Problem 5 (fan power with VFD)

A fan of 3 kW at full speed is slowed to 80 %: power ∝ N^3 → 3 × 0.8^3 = **1.54 kW** (49 % saving) – the reason why VFDs are used.

Short answers (2 marks)

- **Define effective temperature.** Temperature of still saturated air giving the same sensation as the actual air state.
- **What is HEPA?** Filter with ≥ 99.97 % efficiency for 0.3 m particles.
- **Why do forward-curved fans overload?** The power curve keeps rising with flow, so the motor can overload at high Q.
- **Difference between refrigerator and heat pump?** Refrigerator desired effect is Q_L, heat pump desired effect is Q_H; $COP_{HP} = COP_R + 1$.
- **What is ISEER?** Seasonal cooling output / energy input over the Indian cooling season; used for BEE stars.
- **Which humidifier is isothermal?** Steam humidifier.
- **What is the function of eliminator plates?** To prevent water droplets from carrying over into the duct.

Accessories in air-conditioning ducts

- **Dampers** (volume control, fire, smoke dampers), **vibration isolators / flexible joints** for fans, **silencers / acoustic lining** (noise control), **turning vanes** in bends, **grilles / diffusers**, **access doors**, **air curtains** at entrances.

8. Quick Revision and Exam Questions

- Humidifier: spray (adiabatic), steam (isothermal), ultrasonic. Dehumidifier: cooling coil, silica gel / LiCl, desiccant wheel.
- Filters: dry, viscous, wet, ESP, HEPA (99.97 % at 0.3 m), carbon.
- Fan laws: $Q \propto N$, $\Delta P \propto N^2$, $P \propto N^3$; $P = Q\Delta P/\eta$.
- $M - W = E + R + C + S$; ET = still saturated air with the same feeling; comfort ≈ 22-26 °C, 40-60 % RH.
- $COP_{HP} = COP_R + 1$; sources: air, water, ground, solar, waste heat.
- ISEER: higher is better; 5-star split ≥ 5.6 (2026-28).

Likely questions

- Explain different types of humidifiers and dehumidifiers with neat sketches. (long)
- Types of air filters – working and applications; HEPA, ESP. (long)
- Classify fans and blowers; fan laws; fan characteristic curves. (long)
- Define effective temperature; explain the comfort chart; factors affecting comfort. (long)
- Heat pump – working, sources, circuits; compare with refrigerator. (long)
- Energy efficient AC – star rating, ISEER and the features that improve efficiency. (short / long)